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# Intelligent traffic signal control based on reinforcement learning: a survey

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**Abstract**
Rapid urbanization and the surge in vehicle ownership have exacerbated traffic congestion, posing substantial economic, environmental, and social challenges. Traditional traffic signal control methods often struggle to address the dynamic complexities of modern urban traffic, frequently resulting in operational inefficiencies. Reinforcement Learning (RL), with its inherent capacity for real-time learning and adaptation, has emerged as a promising paradigm for optimizing Traffic Signal Control (TSC). RL approaches are particularly well-suited for handling complex traffic states and coordinating global optimization across multiple intersections. Despite notable progress, RL-based systems continue to face significant hurdles, including high computational costs, extensive data requirements, and issues regarding generalizability across diverse traffic scenarios. This paper synthesizes current RL-based models for TSC and highlights recent advancements in the field. It provides a comprehensive review of prominent approaches, categorizes existing studies based on their methodological frameworks, and conducts a technical evaluation of classical RL-based methods to assess their performance across varied traffic conditions. Finally, the remaining challenges and potential future directions for RL-based TSC are critically examined.

**Keywords** Multi-agent reinforcement learning · Traffic signal control · Intelligent transportation system · Applied artificial intelligence

## 1 Introduction

Rapid urbanization and the proliferation of vehicle ownership have exacerbated urban traffic congestion, creating severe challenges for public mobility and societal well-being. Beyond the economic costs of wasted fuel and time, congestion significantly increases hazardous emissions, including greenhouse gases and particulate matter, posing risks to human health. Consequently, effective traffic congestion mitigation has become an urgent priority. Traffic Signal Control (TSC) plays a pivotal role in urban traffic management, and optimizing sig-

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nal timings has been shown to improve ground traffic efficiency by 10–20% Eom and Kim (2020); Qadri et al. (2020); Louati (2020); Chin et al. (2004); Yau et al. (2017).

TSC systems are essential for the intelligent scheduling of traffic flow, offering the potential to reduce accident rates, conserve energy, and lower pollutant emissions. Ideally, these systems dynamically adjust signal timings in response to real-time variations, for example, extending green phases on arterial roads during peak hours or shortening cycle lengths during off-peak periods. However, traditional TSC methods, which often rely on fixed-time or simple actuated logic, lack the flexibility to adapt to the highly dynamic and complex nature of modern urban traffic. As traffic volumes surge, these conventional systems are increasingly insufficient to meet control demands.

In response to these limitations, Reinforcement Learning (RL) Sutton and Barto (1998) has emerged as a promising paradigm for TSC. In an RL formulation, an intersection is commonly modeled as a Markov Decision Process (MDP), where traffic conditions constitute the state, signal control decisions define the action space, and mobility objectives (e.g., delay reduction) are encoded as rewards. This formulation enables controllers to learn adaptive strategies from interaction data rather than relying solely on pre-specified rules. In particular, Deep Reinforcement Learning (DRL) Mnih et al. (2015) leverages deep neural networks as function approximators to handle high-dimensional observations, which has accelerated progress in large-scale and complex TSC settings.

The evolution of RL-based TSC research has progressed through several distinct phases, moving from isolated control to complex, large-scale coordination. The application of RL in this domain traces its origins to the 1990 s, with pioneering works by Mikami and Kakazu (1994) and Thorpe and Anderson (1996) exploring genetic reinforcement learning and SARSA for traffic control. Building on these foundations, the field advanced significantly in the 2010 s as researchers tackled larger networks. For instance, El-Tantawy et al. (2013) proposed the MARLIN-ATSC method, while Wei et al. (2018) later introduced Independent Q-learning (IQL) to learn adaptive policies with rich state representations. However, these early modern approaches often treated intersections as independent agents. Recognizing the necessity of coordination in road networks, subsequent research shifted towards Multi-Agent Reinforcement Learning (MARL) to optimize regional objectives. Approaches like MA2C Chu et al. (2019) and CoLight Wei et al. (2019) incorporated explicit coordination mechanisms to facilitate information exchange and cooperation among intersections.

To further enhance decision-making precision, researchers began integrating deep learning architectures within RL frameworks to capture spatiotemporal dependencies. For instance, DynSTGAT Wu et al. (2021) and MNLight Kong et al. (2022) leverage historical data and recurrent structures to improve state representation. Similarly, Transformer Light Wu et al. (2023) introduces Transformer-based temporal modeling to handle non-stationary patterns more effectively. Concurrently, addressing scalability in large-scale networks became a priority. Methods such as FMA2C Ma and Wu (2020), GPLight Liu et al. (2023), and ReginLight Gu et al. (2024) employ hierarchical learning and task decomposition, such as region partitioning, to manage system complexity and maximize network-level performance. Most recently, the focus has expanded to generalization and transfer across cities and scenarios. Frameworks such as Dualight Lu et al. (2023), CrossLight Sun et al. (2024), X-Light Jiang et al. (2024), and CoSLight Ruan et al. (2024) aim to improve robustness to distribution shifts by combining transferable representations with scenario-specific adaptation.

Despite these advancements, the transition of RL from research prototypes to real-world TSC deployment is impeded by obstacles that can be categorized into three distinct dimensions: data efficiency, perceptual uncertainty, and systemic scalability. The first hurdle involves the sim-to-real gap; the high cost of real-world data collection necessitates a reliance on simulators, which often fail to capture the nuances of physical environments. The second challenge concerns perceptual uncertainty and environment dynamics. Unlike idealized simulations, real-world traffic states are often partially observable or corrupted by sensor noise, requiring the robust integration of RL with dynamic state estimation and filtering techniques to ensure reliable control. Furthermore, the stochastic and non-stationary nature of traffic flows complicates stable convergence. Finally, operational and safety constraints present significant deployment barriers. As networks expand, the exponential growth of state-action spaces threatens real-time responsiveness, while the "black-box" nature of neural policies lacks the interpretability and safety assurances required for critical infrastructure.

Several surveys have reviewed RL for traffic signal control from complementary perspectives. Miletić et al. (2022) focus on RL applications within adaptive traffic signal control (ATSC), covering TSC fundamentals and discussing the implications of connected and autonomous vehicles. Noaeen et al. (2022) present a systematic and reproducible review restricted to network-level TSC studies (two or more intersections), with a comprehensive meta-analysis of methods, environments, and evaluation practices up to March 2020. Wei et al. (2021) survey recent RL models for TSC and summarize commonly used simulation environments and experimental settings for evaluation. Building on and extending these efforts, our survey distinguishes itself in two aspects. First, we provide a unified, implementation-oriented taxonomy that links problem formulation choices (state, action, and reward design) to algorithmic families for both single-intersection and multi-intersection control, enabling a clearer mapping from design decisions to expected behavior and limitations. Second, beyond summarizing prior work, we conduct a comparative experimental evaluation of classical and representative RL-based methods under multiple traffic scenarios and metrics, offering an evidence-based discussion of their strengths, failure modes, and generalization challenges. This combination is intended to complement existing reviews by connecting methodological categorization with reproducible empirical insights.

Motivated by these gaps, this paper provides a comprehensive review of RL-based TSC methods, categorizing existing approaches, assessing their technical performance, and analyzing future directions. To ensure a systematic review, we adopted a structured methodology for literature selection. We performed keyword searches using combinations of terms such as "Traffic Signal Control," "Reinforcement Learning," "Deep Reinforcement Learning," and "Multi-Agent Systems" across major academic databases, including IEEE Xplore, ACM Digital Library, Google Scholar, and Web of Science. The search covered the period from 1980 to 2025, aiming to capture the rapid advancements following the resurgence of deep learning. The initial pool of papers was subsequently filtered based on three criteria: (1) *Relevance:* The study must explicitly apply RL or DRL techniques to TSC problems; (2) *Quality and Impact:* Priority was given to articles published in top-tier journals and conferences, as well as high-impact preprints; (3) *Reproducibility:* For the experimental comparison in Section 5, we specifically selected methods with open-source implementations or detailed reproducibility guidelines. An overview of the paper is presented in Figure 1.

The main contributions of this survey are summarized as follows:

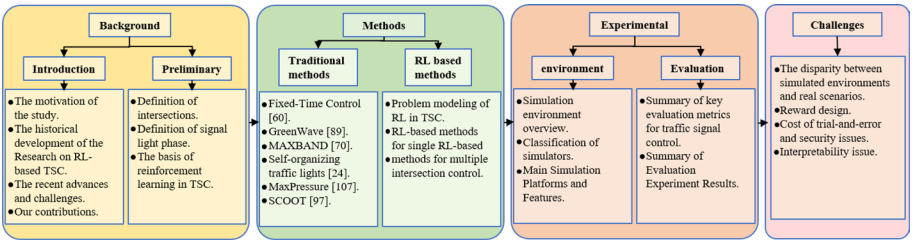



**Fig. 1** Overview of the paper, illustrating the scope of this survey, the taxonomy of traffic signal control (TSC) methods, and the workflow from problem formulation to experimental evaluation and discussion of open challenges

- We provide a comprehensive and systematic review of RL-based TSC methods, categorizing the literature to highlight the evolution from independent agents to collaborative, hierarchical, and transfer-learning-based frameworks.
- We conduct an extensive experimental evaluation of representative RL-based models across multiple datasets, using diverse metrics to ensure a rigorous and objective assessment of current capabilities.
- We critically analyze the limitations of existing studies, focusing on computational complexity, scalability, data efficiency, and the generalization gap between simulation and deployment.
- We identify emerging trends and propose future research directions aimed at bridging the gap between theoretical RL models and practical, large-scale traffic management systems.

The remainder of the paper is organized as follows. Section 2 outlines the problem definition and RL fundamentals. Section 3 reviews traditional TSC benchmarks. Section 4 details RL-based methods for both single and multi-intersection scenarios. Section 5 presents the experimental analysis. Section 6 discusses open challenges, and Section 7 concludes the paper.

## 2 Preliminary

This section provides a concise overview of the fundamental concepts underlying RL-based traffic signal control (TSC). We outline the core elements of TSC problems, including intersections and signal phases, and briefly introduce how RL adapts to this context. Key components such as the simulation environment and common evaluation metrics are also described. For readers interested in detailed definitions of standard terms, we provide a glossary in A (Appendix A).

## 2.1 Definition of TSC problems

### 2.1.1 Definition of intersections

Intersections are the most common and crucial component of the urban transportation environment. The traffic signal control environment is defined as a standard four-leg intersection, formed by the crossing of two orthogonal roads. The intersection accommodates traffic flows from four directions: east, west, south, and north. Each approach is equipped with a set of dedicated lanes based on actual traffic demands. A typical configuration includes designated lanes for left-turn, through, and right-turn movements, as illustrated in the example shown in Fig. 2. The number and functional allocation of lanes can be adjusted according to relevant roadway design standards. Each approach is equipped with a coordinated traffic signal system, and vehicles are required to comply with the signal indications corresponding to their approach lanes for safe passage through the intersection.

When studying the TSC problems, this paper primarily focuses on the vehicle states immediately before entering the intersection, which fall within the observation range of the intelligent agent's perception of the environment. The traffic flow at an intersection refers to the difference between the number of vehicles exiting the intersection and the number of vehicles entering it per unit time. The traffic flow through intersections is typically influenced by various factors, such as a notable increase in traffic during rush hours. However, during sudden adverse weather conditions and nighttime, the traffic flow tends to be relatively low.

### 2.1.2 Definition of signal light phase

Traffic signal lights are generally used at intersections to control the passage of vehicles by generating different traffic signals. An intersection typically has four traffic lights to control vehicle traffic in the east, south, west, and north directions. Each traffic light has three signals: red for stop, yellow for proceed with caution, and green for go. Traffic lights mainly control the left turn and straight ahead of vehicles, and on the right turn lane, vehicles are allowed to turn right by default.

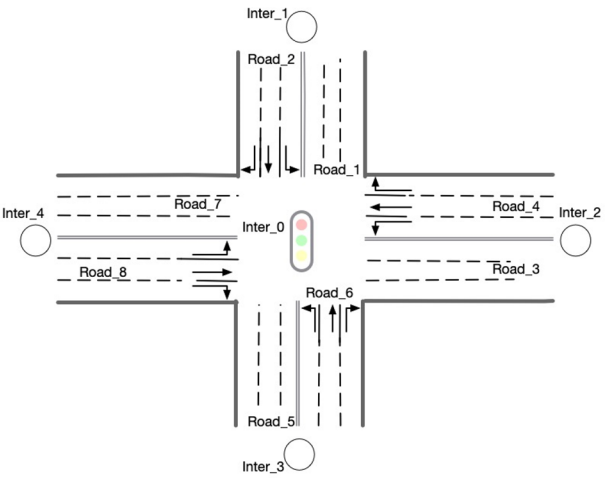



**Fig. 2** Schematic intersection environment used for RL-based TSC, showing incoming and outgoing approaches, lane groups, and traffic streams

For intersections, a set of signal lights that effectively control the traffic flow of vehicles at the intersection usually have green phases in several directions and red phases in other directions. To facilitate a simplified yet effective definition of intersection phases, this study decomposes the intersection control into a set of extensible basic signal phases. As illustrated in Fig. 3, a typical four-phase signal control scheme consists of the following phase groups: (1) east–west through movement; (2) east–west left-turn movement; (3) north–south through movement; and (4) north–south left-turn movement. In practical traffic systems, non-conflicting phases can be combined into composite phases to improve efficiency. For instance, the simultaneous release of east–west through and left-turn movements can result in an eight-phase configuration or even more complex phase schemes. Based on the current traffic flow conditions at the intersection, the agent selects the optimal action to maximize the reward, thereby enhancing vehicle throughput at the intersection.

## 2.2 Reinforcement learning for traffic signal control

RL formulates TSC as a sequential decision-making problem. Instead of relying on predefined rules, an RL agent learns an optimal policy $\pi$ by interacting with the traffic environment modeled as a Markov Decision Process (MDP).

### 2.2.1 Problem formulation

In the context of TSC, the standard MDP tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ is mapped as follows:

- *State ($\mathcal{S}$):* Represents traffic conditions, such as queue lengths, vehicle positions, or image-based snapshots of the intersection.
- *Action ($\mathcal{A}$):* Corresponds to signal control decisions, such as switching to a specific phase or extending the current green time.
- *Reward ($R$):* A feedback signal quantifying traffic efficiency, typically defined as the negative of total delay, queue length, or waiting time.

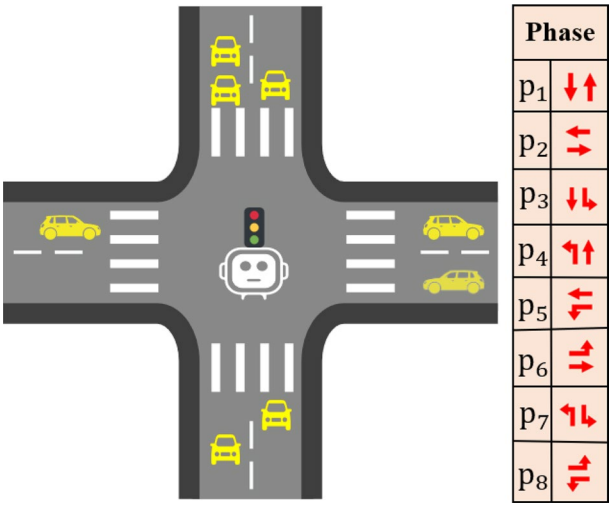



**Fig. 3** Example of an 8-phase signal plan. The figure illustrates feasible phase combinations and non-conflicting movements, which constrain the action space for phase switching or phase selection in RL-based controllers

At each time step $t$, the agent observes state $s_t$, executes action $a_t$, and receives reward $r_{t+1}$. The goal is to maximize the expected cumulative return $G_t$. The interaction loop is illustrated in Fig. 4. For readers interested in the detailed mathematical definitions of value functions and Bellman equations, we provide a reference in Appendix B.

### 2.2.2 Deep reinforcement learning (DRL) adaptations

Traditional tabular RL struggles with the high-dimensional state spaces typical of complex traffic intersections. DRL addresses this by using deep neural networks (DNNs) as function approximators. Key DRL algorithms are adapted for TSC as follows:

*Discrete Control with DQN.* The Deep Q-Network (DQN) Mnih et al. (2013) is widely used for *phase selection* tasks where the action space is discrete (e.g., choosing one of eight phases). As shown in Fig. 5, the DNN maps raw traffic states (e.g., vehicle position matrices) directly to Q-values, enabling the agent to handle complex visual or structured data that traditional sensors cannot easily capture.

*Continuous Control with DDPG.* While DQN handles phase switching, traffic control often requires continuous variables, such as calculating the precise duration of a green light. The Deep Deterministic Policy Gradient (DDPG) Lillicrap et al. (2015) utilizes an Actor-Critic architecture to output continuous action values, allowing for fine-grained adjustments of signal timing that discrete methods cannot achieve.

*Efficient Sampling with A3C.* Traffic simulations can be computationally expensive. The Asynchronous Advantage Actor-Critic (A3C) Mnih et al. (2016) algorithm addresses this by deploying multiple parallel worker threads (Fig. 6). In TSC, this allows agents to collect experiences from diverse traffic scenarios simultaneously, stabilizing training and significantly reducing the wall-clock time required to learn robust policies.

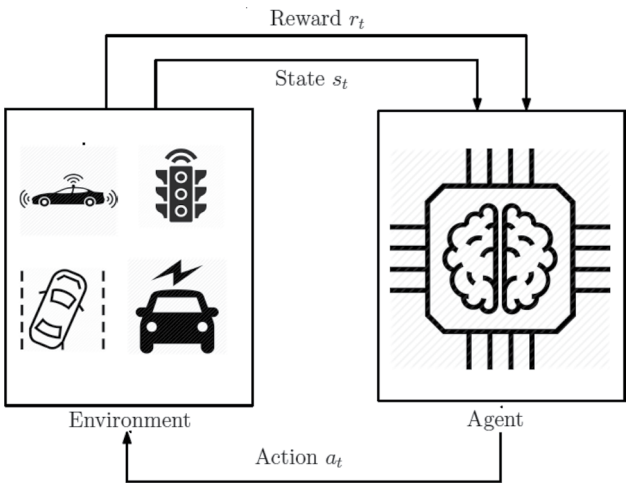



**Fig. 4** Agent–environment interaction loop for RL-based TSC

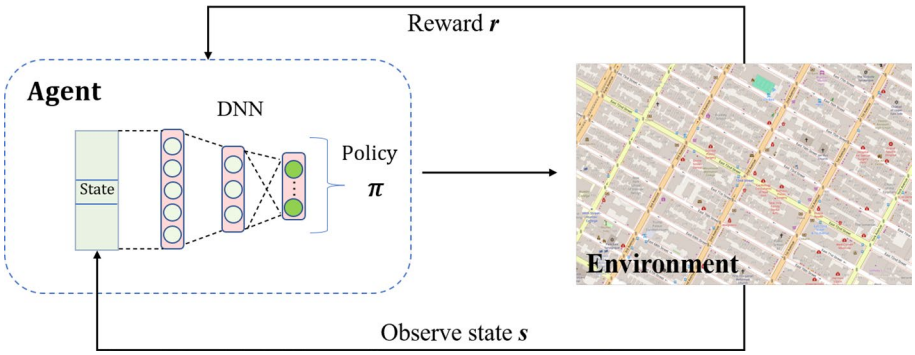



**Fig. 5** Schematic of a deep reinforcement learning agent for traffic signal control, where a DNN maps the observed state to a control policy $\pi$ that interacts with the traffic environment

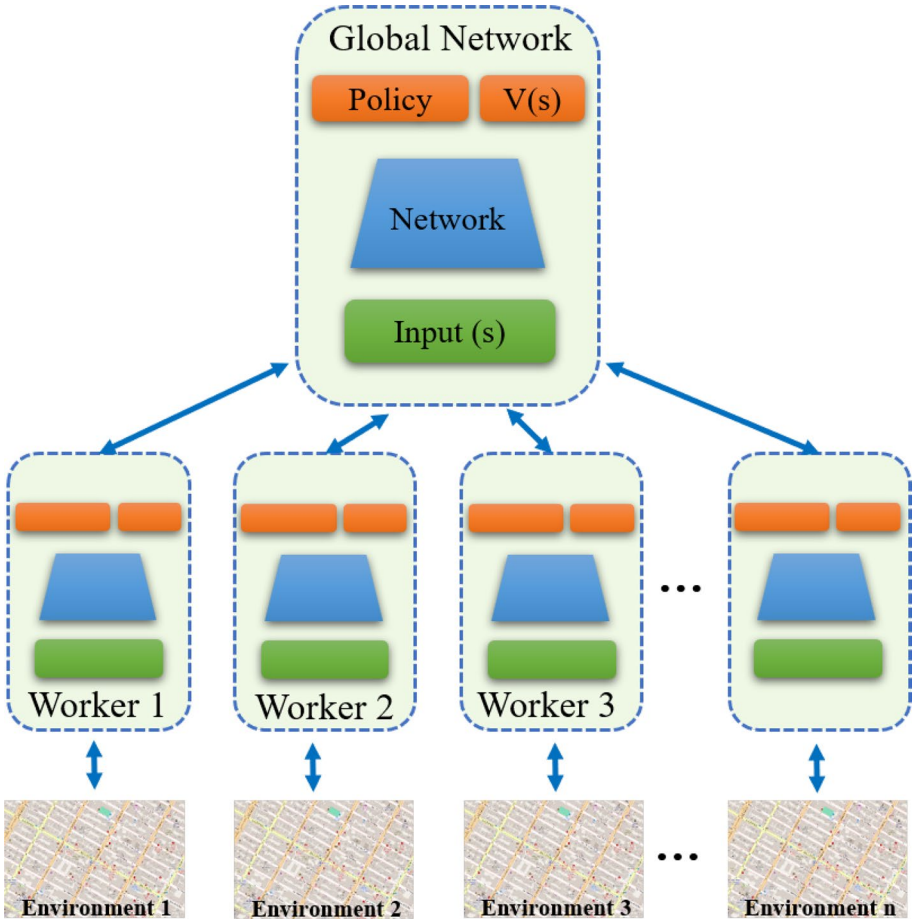



**Fig. 6** Schematic of the Asynchronous A3C framework for TSC. Multiple parallel workers interact with their environments to collect trajectories, and asynchronously update shared actor and critic parameters, enabling faster training and improved exploration compared with single-thread learning

### 2.2.3 Multi-agent reinforcement learning (MARL) for networks

Single-agent RL is insufficient for large-scale road networks due to the "curse of dimensionality" and the lack of coordination. MARL extends the framework to multi-intersection control (Fig. 7), where each intersection acts as an autonomous agent.

In this setting, the environment becomes *non-stationary* from a single agent's perspective because other agents are simultaneously updating their policies. To address this, MARL methods in TSC typically fall into two categories:

- *Joint Action Learning:* A centralized approach that learns a global policy. While theoretically optimal, it scales poorly because the joint action space typically grows exponentially with the number of agents.
- *Independent Learning with Communication:* Agents learn decentralized policies based on local observations but exchange information (e.g., outgoing traffic flow or intended actions) with neighbors. This approach balances scalability with the need for coordination, making it the dominant paradigm for city-scale traffic control.

## 2.3 Simulation environment

Due to the danger of real-world experiments, the application of RL and DRL in TSC is mainly carried out on simulators. Traffic simulators are divided into three categories based on granularity: micro, macro, and meso. Microscopic simulator with a time step of seconds, detailed simulation of vehicle behavior, suitable for fine traffic control and optimization. The macro simulator has a time step of minutes and focuses on the overall behavior of traffic flow, making it suitable for urban transportation planning and policy evaluation. The time step of the mesoscopic simulator is between seconds and minutes, and combined with the

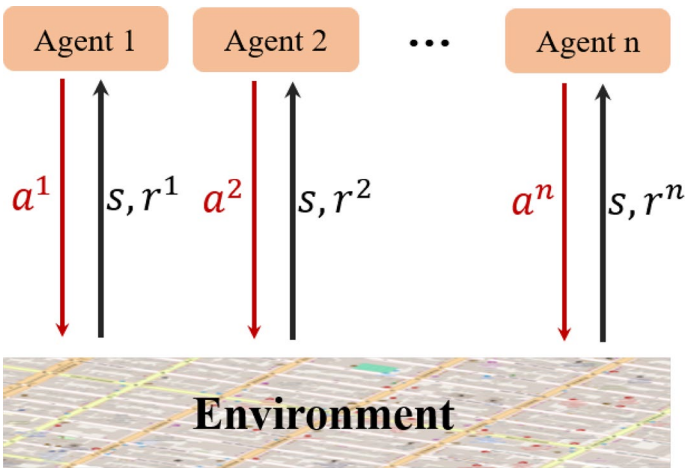



**Fig. 7** Schematic of Multi-Agent Reinforcement Learning for multi-intersection control. Each intersection is modeled as an agent that learns a local policy from local observations, while coordination is achieved via shared objectives, information exchange, or centralized training with decentralized execution to improve network-wide performance

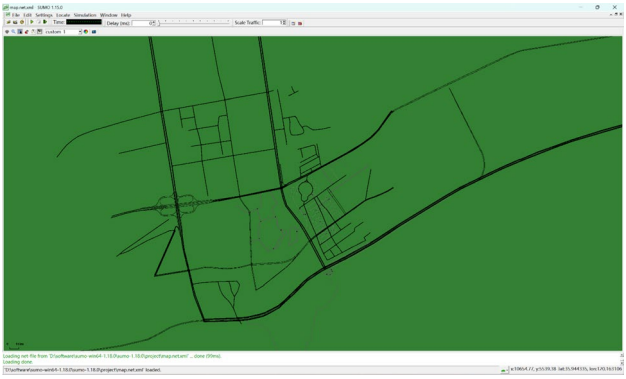


**Fig. 8** SUMO simulator interface used to generate microscopic traffic dynamics for training and evaluating TSC policies, supporting realistic vehicle behavior, detector configurations, and road network modeling

characteristics of both, it is suitable for regional traffic management and signal optimization. Typical traffic simulation environments are as follows:

- *SUMO(Simulation of Urban Mobility)*[1]: an open-source traffic simulation software that provides versatile, multi-scale simulation capabilities, ranging from microscopic to macroscopic levels. It is specifically designed to support urban mobility research and accommodates diverse transportation modes, including automobiles, public transit, and pedestrian movement. Widely employed in traffic engineering, urban planning, autonomous systems development, and multi-agent systems research, SUMO serves as a powerful tool for implementing and evaluating traffic management strategies. It excels in handling large-scale traffic networks and generates detailed simulation data, enabling comprehensive traffic analysis and system development. Furthermore, SUMO includes a Python-based interface, TraCI, which allows users to interact with the simulation environment and seamlessly incorporate reinforcement learning methodologies. The specific interface is shown in the figure 8
- *VISSIM:* a microscopic traffic simulation software that uses time-step and driving behavior models to simulate individual vehicle behavior with high precision. It supports a wide range of traffic modes, including cars, public transportation, pedestrians, and cyclists. VISSIM is renowned for its advanced traffic signal control capabilities, such as adaptive signal control and multi-period signal timing schemes. Users can integrate external programs like VAP or VSPLUS to implement dynamic signal control, enhancing flexibility in traffic management. Additionally, VISSIM offers a COM interface for Python-based secondary development, enabling users to access simulation states and customize signal control strategies with ease.
- *GLD (Green Light District):* a microscopic traffic simulation software based on cellular automata, designed primarily to evaluate the performance of AI-driven traffic signal control algorithms. It simulates individual vehicle behaviors, including position, speed, and acceleration, delivering high-precision traffic flow analysis. GLD is widely employed for testing and optimizing traffic signal control algorithms, particularly in multi-

[1] https://www.eclipse.org/sumo/

agent reinforcement learning contexts, where it enables the simulation of collaboration between adjacent traffic lights. With its support for large-scale traffic network simulations, GLD is well-suited for urban traffic management and optimization research. However, it faces challenges in computational efficiency and realism when applied to large and complex traffic scenarios.

- *Aimsun:* a widely used microscopic traffic simulation software developed by TSS (Transport Simulation Systems) in Spain, extensively applied in traffic engineering and planning. It provides multi-scale simulation capabilities, enabling the modeling of complex traffic scenarios involving vehicles, pedestrians, and public transportation. Aimsun simulates individual vehicle behaviors in detail, including acceleration, deceleration, lane-changing, and car-following dynamics, using advanced vehicle-following and lane-changing models to deliver highly realistic traffic flow simulations. It is renowned for its robust traffic signal control features, supporting adaptive signal control and multi-period signal timing schemes. By integrating with external programs like Python or MATLAB, users can implement dynamic signal control strategies to optimize traffic signal timings. Furthermore, Aimsun facilitates coordinated signal control across multiple intersections, improving overall network efficiency.
- *Paramics:* a widely used microscopic traffic simulation software developed by Quadstone Paramics in the UK, extensively applied in traffic engineering and management. It supports multi-scale simulation capabilities, enabling the modeling of complex traffic scenarios. Paramics provides a Python interface that allows users to access simulation states and implement custom functionalities through scripts. Furthermore, it facilitates the integration of reinforcement learning algorithms, such as Q-learning and DQN, enabling the optimization of traffic signal control and autonomous driving strategies.
- *AIM (Autonomous Intersection Management):* a project developed by the AI Laboratory at the University of Texas at Austin, designed to create a scalable, safe, and efficient intersection management environment. The AIM simulator allows for user-defined intersection layouts and customizable vehicle behavior models, enabling the simulation of complex traffic scenarios to address diverse research needs.
- *CityFlow*[2]: an open-source traffic simulation platform designed to provide multi-scale simulation capabilities for vehicular traffic in urban networks. It supports large-scale traffic scenario simulations and offers flexible interfaces, making it ideal for research areas such as traffic signal control and autonomous driving path planning. Leveraging multi-threaded parallel computing and optimized data structures, CityFlow achieves high simulation efficiency, operating over 20 times faster than SUMO. It is capable of simulating urban traffic networks with hundreds of intersections and tens of thousands of vehicles. The platform supports customizable road network and traffic flow configurations, enabling simulations based on synthetic or real-world data. Furthermore, it features a user-friendly Python interface that facilitates integration with reinforcement learning algorithms, including multi-agent reinforcement learning. The specific interface is shown in the figure 9

[2] https://cityflow-project.github.io/

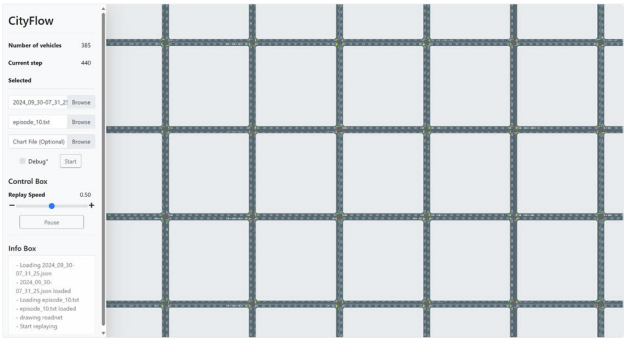



**Fig. 9** CityFlow simulator interface for large-scale, high-efficiency traffic simulation. CityFlow is commonly used for benchmarking RL-based TSC under city-level networks due to its computational scalability

## 2.4 Evaluation metrics

The evaluation of traffic control systems, particularly those utilizing reinforcement learning (RL) methods, relies on several key performance metrics. These metrics offer valuable insights into the efficiency and effectiveness of various control strategies, including both multi-intersection and single-intersection RL approaches, as well as traditional traffic management systems.

- *Average Travel Time(ATT):* This metric represents the mean time taken by vehicles to travel from entry to exit within a given road network or traffic system. It encompasses the entire journey, including delays caused by red lights, slow driving, and traffic congestion. A shorter average travel time indicates a more efficient traffic system and a well-performing traffic signal control strategy, enabling vehicles to traverse the network more quickly.
- *Throughput(TP):* This metric refers to the total number of vehicles that successfully pass through a road network or specific sections (such as intersections, segments, or the entire system) within a specified time period. This metric reflects the network's capacity and efficiency in managing traffic flow. High throughput signifies an effective traffic signal control strategy that mitigates congestion, manages peak traffic flow, and enhances travel efficiency.
- *Average Delay Time(ADT):* This metric quantifies the additional waiting time experienced by vehicles when passing through a road network or specific sections due to traffic signals or other management measures. It evaluates the efficiency of traffic control strategies by highlighting the extent of delays and their impact on travel efficiency.
- *Average Queue Length(AQL):* This metric measures the number of vehicles waiting in line to pass through a road network or specific sections within a given time period. It serves as an indicator of congestion and the effectiveness of traffic management strategies.
- *Average Waiting Time(AWT):* This metric refers to the duration vehicles spend stationary due to traffic signals or other management measures when navigating through a road network or specific sections. This metric provides a focused assessment of the delays

imposed by traffic control systems.

- *Average Rewards(AR):* In RL-based traffic signal control systems, this metric represents the mean rewards earned by the control strategy or algorithm over a specified time period. Average rewards reflect the effectiveness of the system in optimizing traffic flow, reducing congestion, and improving efficiency. Higher average rewards indicate a well-designed strategy that successfully minimizes waiting and travel times, ultimately enhancing the overall performance of the traffic system.

## 3 Traditional traffic signal control methods

This section reviews representative traditional traffic signal control (TSC) methods. These approaches provide essential baselines for RL-based controllers and help clarify how signal control has evolved from offline planning to online adaptation. Following standard classifications in transportation engineering, we group traditional TSC into three categories: (i) fixed-time (pretimed) control, (ii) actuated control, and (iii) adaptive control. Figure 10 summarizes this taxonomy and lists representative examples.

### 3.1 Fixed-time (pretimed) control

*Fixed-Time Control* Koonce et al. (2008). Fixed-time control computes a signal plan offline from historical demand, including a predetermined phase sequence, cycle length, and green splits. It is straightforward to deploy and can perform well under stable and recurrent traffic patterns. However, because the plan is not adjusted online, performance can deteriorate under incidents, demand surges, or other non-recurrent conditions.

*GreenWave* Mcshane et al. (1998). GreenWave is a coordination strategy designed primarily for arterial corridors. By choosing appropriate offsets among adjacent intersections, it enables platoons to traverse successive intersections with fewer stops at a target cruise speed. A basic offset can be approximated by

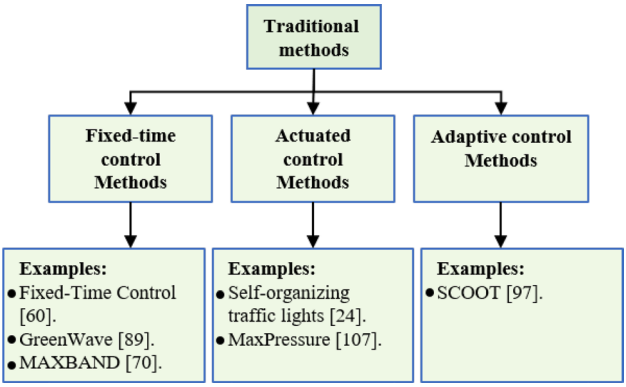



**Fig. 10** Taxonomy of traditional traffic signal control methods. The figure summarizes representative fixed-time (pretimed), actuated (self-organizing), and adaptive/optimization-based approaches, providing baseline categories against which RL-based methods are compared in this survey

$$t_{i,j} = \frac{d_{i,j}}{v}, \tag{1}$$

where $t_{i,j}$ denotes the offset between intersections $i$ and $j$, $d_{i,j}$ is the distance between them, and $v$ is the desired progression speed. GreenWave is easy to implement and can reduce stops and delay along the coordinated direction. Its limitations include sensitivity to speed variability and demand asymmetry, potential adverse impacts on cross-street traffic, and reduced effectiveness in irregular networks or under rapidly changing demand.

*MAXBAND* Little et al. (1980). MAXBAND formulates arterial coordination as an optimization problem that maximizes the bandwidth of green progression. It typically applies mixed-integer linear programming to determine offsets and cycle length subject to operational constraints. MAXBAND yields interpretable progression plans and can be effective for corridor operations. Nevertheless, it may be computationally burdensome for large networks and can be less robust when real-world traffic deviates from assumed demand patterns.

### 3.2 Actuated control

Actuated control updates signal indications online using detector inputs, such as vehicle presence, queue occupancy, or gap times, rather than strictly following a fixed schedule. Compared with pretimed control, actuated strategies can better accommodate short-term demand fluctuations. However, many actuated controllers rely on local rules and therefore may not explicitly optimize network-level objectives.

*Self-Organizing Traffic Lights (SOTL)* Cools et al. (2013). SOTL is a decentralized actuated strategy that switches phases using locally observed traffic measurements, such as queue length and waiting time. For example, when the accumulated demand on an approach exceeds a threshold, the controller serves that movement. SOTL is lightweight and does not require a central coordinator, making it suitable for dynamic settings. Its main limitation is that decisions are driven by local information, which can be insufficient for achieving strong corridor progression or network-level coordination under highly unbalanced flows.

*MaxPressure as a self-organizing (actuated) controller* Varaiya (2013). MaxPressure is a representative pressure-based policy within the broader family of self-organizing actuated control methods. It selects the phase that maximizes instantaneous pressure release, where pressure is typically defined by queue imbalances between upstream incoming links and downstream outgoing links. MaxPressure is attractive due to its decentralized implementation, reliance on real-time measurements, and strong throughput properties under standard assumptions. In practice, however, because it optimizes a proxy objective at each decision step, it can yield suboptimal outcomes under multi-criteria goals such as fairness, spillback prevention, or corridor progression.

### 3.3 Adaptive control

Adaptive control systems continually adjust timing parameters using online measurements and explicit optimization objectives, typically at the corridor, subnetwork, or network level. In contrast to actuated control, adaptive approaches generally incorporate model-based esti-

mation and repeated re-optimization of key parameters such as cycle length, splits, and offsets.

*SCOOT* Stevanovic and Martin (2008). SCOOT is a widely deployed adaptive signal control system that updates timing parameters using detector data. In general, SCOOT adjusts (i) cycle length, (ii) green splits, and (iii) offsets to reduce delay and stops. A common representation is $C = C_{\text{base}} + \Delta C$ for cycle length and $G_i = G_{\text{base},i} + \Delta G_i$ for green time allocation, where the increments are updated based on observed traffic conditions. Offsets are also adjusted to maintain progression along key routes. Although SCOOT has demonstrated effectiveness in practice, it requires substantial sensing infrastructure and calibration, leading to high installation and maintenance costs. Moreover, performance depends on model fidelity and timely parameter updates, which introduces a practical trade-off between model complexity and real-time responsiveness.

## 4 RL based traffic signal control methods

In recent years, RL and DRL have achieved notable progress and delivered strong performance across many domains. In traffic signal control (TSC), RL-based approaches have likewise attracted considerable interest. This section introduces representative RL-based control methods for TSC. We provide an overview in Figure 11. Given that RL methods require an explicit problem formulation, we first describe how TSC is modeled in an RL framework.

### 4.1 Problem modeling of RL in TSC

#### 4.1.1 Definition of state for RL in TSC

In reinforcement learning, state representation is a core concept that describes the situation of the environment at a specific moment in time. For traffic signal control problems, state representation usually includes key information about intersections and their surrounding environments. In traffic signal control, various features have been proposed to represent environmental states, as follows:

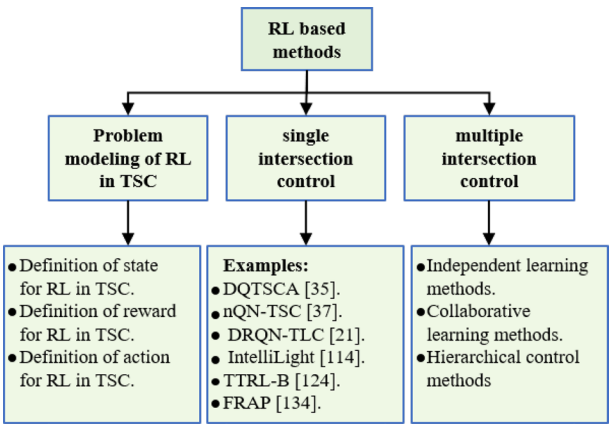



**Fig. 11** Taxonomy of RL-based TSC methods reviewed in this paper. We categorize studies by (i) problem formulation (state, action, and reward design) and (ii) control scope (single-intersection versus multi-intersection control), where multi-intersection methods further include independent, collaborative, and hierarchical coordination paradigms

- Current signal phase: indicates the status of traffic lights in each direction.
- Vehicle queue length: The number of vehicles waiting on each lane.
- Traffic flow: The rate of vehicles passing through in all directions.
- Waiting time: The average waiting time of vehicles on each lane.
- Vehicle speed: Distribution of vehicle speeds near intersections.
- Phase duration: The duration of the current phase, which is crucial for deciding whether to switch phases.
- Vehicle location: not only includes queue length, but also the specific distribution of vehicles on the road.
- Delay: The average vehicle delay time in each lane or direction.
- Number of vehicles on different lanes: including the number of vehicles on each segment of each entering lane and the number of vehicles on each exiting lane.

Beyond the selection of specific features, recent advancements have emphasized the reliability of state estimation, particularly in addressing the dynamic nature of urban traffic and sensor noise. Traditional RL approaches often assume perfect state observability; however, real-world deployments require robust integration with filtering techniques. A novel direction involves leveraging the interaction between control actions and state dynamics. For instance, Taher et al. (2025) proposed a filter-based approach for traffic congestion prediction that utilizes dynamically interchangeable state transition matrices aligned with specific traffic signal control actions. By explicitly incorporating signal phases into the estimation process, this method achieves superior prediction accuracy (lower RMSE) compared to traditional Kalman or particle filters, thereby providing the RL agent with a more robust state representation even under communication failure scenarios.

With advances in computer vision and deep learning, image-based state representations have been increasingly explored in RL for traffic signal control. This approach transforms complex traffic scenes into visual inputs, providing an alternative to handcrafted numerical features. Wei et al. (2018) defined an image-based state representation using vehicle position maps. Compared with discrete features, image representations better preserve the spatial distribution of vehicles. However, they often rely on thousands of image features, which can lead to the curse of dimensionality and degrade learning efficiency as well as real-time performance. Moreover, in high-dimensional image state spaces, RL algorithms may require millions or even billions of environment interactions to converge to a reasonable policy, which increases training time and computational cost and makes online learning difficult in real-world traffic environments Pol and Oliehoek (2016).

#### 4.1.2 Definition of reward for RL in TSC

In reinforcement learning, the design of the reward function is crucial for guiding the agent toward effective strategies. Contemporary research employs various metrics to define rewards for traffic signal control. Common reward formulations include:

1. *Minimization of waiting time:* This reward function optimizes traffic flow by minimizing the weighted waiting time of all vehicles. A weight factor is assigned to each vehicle to prioritize specific categories, such as emergency vehicles or public transportation:

$$r_w = -\sum_i w_i \cdot t_i \tag{2}$$

where $w_i$ represents the weight of the *i*-th vehicle, and $t_i$ denotes the accumulated waiting time of the *i*-th vehicle.

2. *Maximizing throughput:* This objective aims to maximize the number of vehicles traversing the intersection per time step, directly reflecting the system's processing capacity:

$$r_t = \sum_i v_i \tag{3}$$

where $v_i$ represents the count of vehicles passing through the intersection at the current time step.

3. *Minimizing queue length:* Focusing on preventing congestion, this function minimizes the queue length across all lanes. It is particularly effective for balancing traffic distribution during peak hours:

$$r_q = -\sum_i q_i \tag{4}$$

where $q_i$ denotes the queue length of the *i*-th lane.

4. *Multi-objective comprehensive reward:* This formulation aggregates multiple performance metrics-waiting time, throughput, and queue length-allowing for a balanced optimization objective tailored to specific requirements:

$$r_m = w_1 \cdot r_{\mathrm{w}} + w_2 \cdot r_{\mathrm{t}} + w_3 \cdot r_{\mathrm{q}} \tag{5}$$

where $r_{\mathrm{w}}$, $r_{\mathrm{t}}$, and $r_{\mathrm{q}}$ correspond to the rewards for waiting time, throughput, and queue length, respectively. The parameters $w_1$, $w_2$, and $w_3$ are weighting coefficients that determine the relative importance of each objective.

5. *Minimizing Pressure:* The pressure $P_i$ quantifies the density imbalance between the incoming and outgoing lanes at intersection *i*. Minimizing this metric facilitates an even distribution of vehicles, thereby optimizing green light utilization and overall throughput:

$$r_i = -P_i \tag{6}$$

The pressure $P_i$ is defined as:

$$P_i = \left| \sum_{(l,m)\in i} w(l,m) \right| \tag{7}$$

Here, $w(l, m)$ is a weighting function representing the imbalance, calculated as:

$$w(l,m) = \frac{x(l)}{x_{\max}(l)} - \frac{x(m)}{x_{\max}(m)} \tag{8}$$

where $x(l)$ and $x(m)$ represent the current vehicle counts on incoming lane $l$ and outgoing lane $m$, respectively, while $x_{\max}(l)$ and $x_{\max}(m)$ denote the maximum vehicle capacities of the respective lanes.

#### 4.1.3 Definition of action for RL in TSC

Existing RL-based TSC studies commonly differ in how they define the action space. Beyond phase-only or timing-only formulations, some methods jointly model phase selection and timing using structured (e.g., multi-discrete) or parameterized actions. In general, action definitions can be grouped into the following categories:

- *Phase-duration selection (timing-only within a fixed phase).* In this setting, the agent selects the duration of the current active phase from a predefined set of candidate green times (or green extensions). The phase itself is assumed to be fixed during the decision step.
- *Cycle-level split (timing-only over a fixed phase plan).* Here, the action specifies the green split ratios (or green times) for the next cycle. Typically, the cycle length and phase set are given, and candidate splits are predefined; the agent allocates green time across phases at the cycle level.
- *Phase switching decision in a fixed phase sequence (phase-only).* In this formulation, the action is a binary decision that determines whether to keep the current phase or switch to the next phase in a predetermined cyclic order.
- *Next-phase selection in a variable phase sequence (phase-only).* When the phase order is not fixed, the agent directly selects the next phase to activate. The action is typically the index of the next phase, which allows more flexible phase sequences than cyclic switching.
- *Joint phase and timing actions (hybrid, multi-discrete, or parameterized).* In this class, the agent determines both *which phase* to serve next and *how long* to serve it. This can be implemented as a multi-discrete action (e.g., choosing a phase index and a duration level simultaneously) or as a parameterized action, where a discrete phase choice is paired with a continuous timing parameter. Such joint formulations provide finer control granularity and can better adapt to heterogeneous and mixed traffic conditions Yang and Fan (2024).

### 4.2 RL-based methods in TSC

This section provides a systematic review of RL-based TSC methods. To structure our analysis, we categorize existing approaches based on the scope of control into two primary groups: *single-intersection control*, where an agent optimizes a local strategy for an isolated junction, and *multi-intersection control*, which focuses on coordinated strategies across a network. As illustrated in Figure 12, the field has evolved significantly over the past decades, progressing from early tabular methods (e.g., Q-learning, SARSA) to sophisticated Deep Reinforcement Learning (DRL) frameworks capable of hierarchical and collaborative decision-making. In the following subsections, we critically examine these models based on their fundamental characteristics, including state and action representations, reward engineering, and neural network architectures.

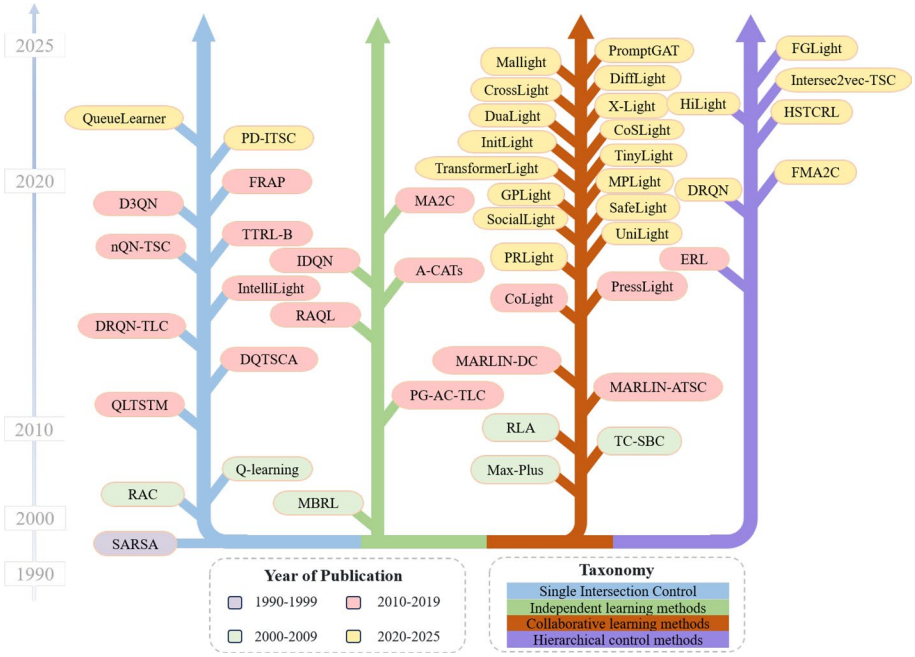



**Fig. 12** Timeline and taxonomy of RL-based Traffic Signal Control methods. The visualization traces the evolution of the field from foundational tabular approaches (1990 s) to modern Deep RL paradigms (2010-2025), categorized by control scope: Single Intersection, Independent, Collaborative, and Hierarchical learning methods

### 4.2.1 RL-based methods for single intersection control

Since the late 1990 s, the application of RL to TSC has evolved from simple tabular methods to sophisticated DRL architectures. In this context, intelligent agents interact with the simulation environment to learn optimal control strategies. We categorize these developments into two primary phases: early tabular methods focusing on feasibility, and modern deep learning approaches addressing high-dimensional state spaces and training stability.

*1. Early Exploration: Tabular RL and Discrete States.* The pioneering work of Thorpe and Anderson (1996) demonstrated the feasibility of RL by applying SARSA to a $4 \times 4$ grid, albeit with limited state representations. Following this, researchers focused on improving state definitions and learning efficiency within tabular frameworks. Abdulhai et al. (2003) and El-Tantawy and Abdulhai (2010) utilized Q-learning with queue lengths and phase times, successfully reducing delays compared to fixed-time controllers. To address the stochastic nature of traffic, Wen et al. (2007) introduced the RAC model with random control mechanisms, while Chin et al. (2011) and Shoufeng et al. (2008) focused on adaptively allocating green time based on queue lengths. Further refinements included Touhbi et al. (2017), who improved congestion representation using queue-to-lane ratios. However, these tabular methods were fundamentally constrained by the "curse of dimensionality," failing to scale when state spaces became complex or continuous.

*2. The DRL Revolution: Handling High-Dimensionality.* The introduction of Deep Reinforcement Learning (DRL) marked a paradigm shift, enabling agents to process raw, high-

dimensional inputs. Genders and Razavi (2016) pioneered this transition with DQTSCA, using Convolutional Neural Networks (CNNs) to map discrete traffic state encodings (DTSE) to actions. This approach was further refined by Gao et al. (2017) and Li et al. (2016), who employed Stacked Autoencoders (SAE) to extract latent features from queue data. To capture the temporal dynamics of vehicle trajectories-which snapshot-based CNNs often Choe et al. (2018) and Shi and Chen (2018) integrated Long Short-Term Memory (LSTM) units into the DRQN architecture, significantly improving performance under partially observable conditions.

*3. Addressing Stability and Convergence.* While DRL offered powerful representations, standard algorithms often suffered from instability. Mousavi et al. (2017) and Garg et al. (2018) addressed this by adopting Policy Gradient methods, which directly optimize the policy space and avoid the value-function bias issues found in Q-learning. Alternatively, Wan and Hwang (2018) introduced dynamic discount factors, and Liang et al. (2019) incorporated Double Dueling DQN (D3QN) with priority replay to stabilize training. Genders and Razavi (2018, 2019) further explored asynchronous training (A3C) and n-step learning to enhance data efficiency.

*4. Domain-Specific Architectures and Mechanisms.* Recent advancements have moved beyond generic networks to architectures tailored for traffic physics. IntelliLight Wei et al. (2018) introduced "phase gate" and "memory palace" mechanisms to handle phase-dependent decision-making. Building on this, Zheng et al. (2019) proposed FRAP, which leverages the symmetry of traffic movements through a "phase competition" mechanism, achieving superior generalization compared to generic models. Similarly, QueueLearner Han et al. (2022) integrates traffic flow theory into the learning process, identifying queue evolution patterns to constrain the RL search space. Other notable innovations include visual-based end-to-end learning by Shabestary and Abdulhai (2018), transfer learning frameworks like TTRL-B Xu et al. (2019) for efficiency, and PD-ITSC Zhang et al. (2020) for handling partial detection scenarios.

Table 1 summarizes these methods, highlighting the evolution of model architectures and their specific contributions to solving TSC challenges.

#### 4.2.2 RL-based methods for multi-intersection control

While single-agent control methods often yield satisfactory local results, they inherently lack communication mechanisms and collaborative decision-making capabilities. This limitation makes it difficult to achieve global optimality in large-scale transportation networks. To address this, Multi-Agent Reinforcement Learning (MARL) has become the standard paradigm. In this section, we classify multi-intersection control methods into three categories based on their coordination strategy: *Independent Learning*, *Collaborative Learning*, and *Hierarchical Control*. We analyze how each category handles the trade-off between scalability and coordination efficacy.

*1. Independent learning methods: Addressing Non-Stationarity*

Independent Reinforcement Learning (IRL) treats each intersection as an autonomous agent. The primary challenge here is *non-stationarity*: since all agents update their policies simultaneously, the environment dynamics change from the perspective of any single agent.

*Early Distributed Approaches.* Wiering et al. (2000) pioneered this field by proposing a distributed Q-learning method with six intersections, utilizing local, mixed, and global

**Table 1** Summary of methods and their contributions in single intersection control

| Method | Algorithm | State representation | Key contribution/mechanism |
|---|---|---|---|
| Thorpe and Anderson (1996) | SARSA | Vehicle counts | First application of RL to TSC |
| RACWen et al. (2007) | SARSA | Sparse discretization | Improved control with random mechanisms |
| Abdulhai et al. (2003) | Q-learning | Queue length | Adaptive control using queue-based states |
| El-Tantawy and Abdulhai (2010) | Q-learning | Queue/Delay | Variable phase sequence control |
| El-Tantawy et al. (2014) | Q-learning | Queue/Delay | Designed reward functions for demand fluctuations |
| QLTSTMChin et al. (2011) | Q-learning | Queue length | Adaptive green light allocation efficiency |
| Shoufeng et al. (2008) | Q-learning | Total delay | Explored green time vs. delay relationship |
| Touhbi et al. (2017) | Q-learning | Queue-to-lane ratio | Improved congestion representation |
| DQTSCA Genders and Razavi (2016) | DQN | Discrete Traffic Encoding | First DRL architecture with discrete encoding |
| Genders and Razavi (2018) | A3C | Image/Vector | Evaluated state representation impact with A3C |
| nQN-TSC Genders and Razavi (2019) | n-step Q-learning | Density/Queue | Asynchronous adaptive control with n-step updates |
| Li et al. (2016) | SAE + Q-learning | Queue length | Stacked Autoencoder for Q-function estimation |
| Gao et al. (2017) | DQN | Position/Speed/Signal | Enhanced representations for reduced latency |
| Mousavi et al. (2017) | Policy Gradient | Raw Image | Addressed training stability via Policy Gradient |
| (Shabestary et al.2018) Shabestary and Abdulhai (2018) | DQN + CNN | Raw Image | Visual-based end-to-end learning framework |
| Garg et al. (2018) | Policy Gradient | Visual features | Real-time dynamic optimization via visual DL |
| DRQN-TLC Choe et al. (2018) | DRQN (LSTM) | Trajectories | Captured temporal dynamics via LSTM |
| Wan and Hwang (2018) | DQN | State-Action | Dynamic discount factors for stability |
| IntelliLight Wei et al. (2018) | DQN | Image + Phase | Phase gate and memory palace mechanisms |
| TTRL-B Xu et al. (2019) | DQN | Transfer samples | Batch-based targeted transfer learning |
| D3QN Liang et al. (2019) | Double Dueling DQN | Image/Phase | Double Dueling architecture with priority replay |
| PD-ITSC Zhang et al. (2020) | Deep Q-learning | Partial states | Addressed partial detection/sensor cost issues |
| FRAP Zheng et al. (2019) | DQN | Phase Pressure | Phase competition for symmetry invariance |
| QueueLearner Han et al. (2022) | DQN | Queue Evolution | Physics-informed queue evolution model |

strategies. Similarly, Camponogara and Kraus (2003) modeled the network as a stochastic game of $N$ agents. To handle congestion propagation, Steingrover et al. (2005) incorporated congestion information from nearby intersections into the local state, while Iša et al. (2006) integrated accident indicator bits to respond to non-recurrent events. While these early methods established the feasibility of distributed control, they often struggled with convergence due to the lack of explicit stabilization mechanisms.

*Multi-Objective and Advanced Reward Design.* To adapt to complex environments, researchers turned to multi-objective optimization. Houli et al. (2010) introduced Multi-RL, which dynamically selects objectives (e.g., waiting time vs. queue length) based on traffic conditions. Brys et al. (2014) combined distributed exploration with reward shaping, while Pham et al. (2013) focused on balancing exploration and exploitation. Khamis et al. (2012); Khamis and Gomaa (2012, 2014) systematically advanced this direction by using Bayesian inference for dynamic objective selection, significantly enhancing performance under adverse weather. Jin and Ma (2019) further structured this using a Master–Slave mechanism to prioritize conflicting objectives.

*Stabilization Mechanisms for Scalability.* Modern IRL methods focus on mitigating non-stationarity without heavy communication. Prashanth and Bhatnagar (2010, 2011) reduced state complexity via feature discretization and introduced synchronous Actor-Critic (PG-AC-TLC) for stable policy evaluation. Aslani et al. (2017) employed tile coding with RBF to handle continuous states. To improve regional coordination, Araghi et al. (2015) used group dimensionality reduction, and Chu et al. (2016) proposed RAQL, which uses grid-based segmentation for regional subnetwork control. Aziz et al. (2018) introduced RMART with standardized queuing indicators to facilitate implicit sharing. Critically, newer methods address agent heterogeneity. Calvo and Dusparic (2018) proposed IDQN with "fingerprinting" to track neighbor policies, while Casas (2017) and Lin et al. (2018) utilized DDPG and PPO with mixed rewards to stabilize continuous control. Finally, MA2C Chu et al. (2019) represents a state-of-the-art baseline, using a decentralized Advantage Actor-Critic (A2C) framework with a "global" field of view, effectively mitigating local myopia by sensing neighbor traffic.

Table 2 summarizes these methods, highlighting the specific mechanisms employed to handle non-stationarity.

*2. Collaborative learning methods: From Message Passing to Graph Networks*

Unlike independent learning, collaborative methods enable agents to explicitly exchange information to optimize a joint objective. The evolution of this field is characterized by a shift from computationally expensive message-passing algorithms to efficient graph-based neural networks.

*Explicit Communication and Coordination.* Early approaches relied on iterative message passing. Kuyer et al. (2008) and Bakker et al. (2010) pioneered the use of the Max-Plus algorithm, allowing agents to negotiate joint actions via coordination graphs. Arel et al. (2010) used a central-peripheral framework, while Taylor et al. (2011) proposed DCEE for distributed exploration. To address dimensionality, El-Tantawy et al. (2013) developed MARLIN-ATSC with modular Q-learning, and Xu et al. (2013) enhanced this with Bayesian estimation (MARLIN-DC). Balaji et al. (2010) proposed RLA to reduce communication complexity via indirect coordination. Later, Pol and Oliehoek (2016) and Shi and Chen (2018) integrated Deep Learning with Max-Plus, utilizing DQN and LSTM to handle state

**Table 2** Summary of independent learning methods (focus: handling Non-stationarity)

| Method | Algorithm | Stabilization mechanism | Key contribution |
|---|---|---|---|
| MBRL Wiering et al. (2000) | Q-learning | None (Naive) | First distributed MARL for TSC |
| Camponogara and Kraus (2003) | Q-learning | Game theory | Modeled network as a stochastic game |
| Steingrover et al. (2005) | Q-learning | Neighbor congestion | Incorporated neighbor states into local view |
| Iša et al. (2006) | Q-learning | Accident indicators | Integrated accident status for robustness |
| Multi-RL Houli et al. (2010) | Q-learning | Multi-objective | Dynamic objective selection for stability |
| Brys et al. (2014) | SARSA | Reward shaping | Distributed exploration via reward shaping |
| Khamis et al. (2012) | Q-learning | Bayesian inference | Dynamic objective selection via Bayesian logic |
| Jin and Ma (2019) | DQN | Master–Slave | Hierarchical priority for multi-objectives |
| Prashanth and Bhatnagar (2010) | Q-learning | Feature discretization | Reduced state space complexity |
| PG-AC-TLC Prashanth and Bhatnagar (2011) | Actor-Critic | Policy gradient | Synchronous policy evaluation |
| A-CATs Aslani et al. (2017) | Actor-Critic | Tile coding | Function approximation for continuous states |
| Araghi et al. (2015) | Q-learning | Group reduction | Regional coordination via dimensionality reduction |
| RAQL Chu et al. (2016) | Q-learning | Region decomp | Real-time traffic flow segmentation |
| RMART Aziz et al. (2018) | Q-learning | Dynamic rewards | R-Markov average reward technique |
| IDQN Calvo and Dusparic (2018) | DQN | Fingerprinting | Heterogeneous agents with fingerprinting |
| Casas (2017) | DDPG | Reward decomp | Continuous control with centralized learning |
| Lin et al. (2018) | ResNet + PPO | Mixed reward | Parallel RL with generalized advantage estimation |
| MA2C Chu et al. (2019) | A2C | Global perception | Decentralized A2C with neighborhood sensing |

complexity while maintaining explicit coordination. Ye et al. (2023) further accelerated training using adversarial inverse RL (InitLight) to imitate expert demonstrations.

*Implicit Graph Coordination (GNNs).* To overcome the latency of explicit messaging, state-of-the-art research has shifted towards Graph Neural Networks (GNNs). Nishi et al. (2018) and Devailly et al. (2021) (IG-RL) first applied GCNs to extract road topology features. However, the breakthrough came with Attention mechanisms. CoLight Wei et al. (2019) utilized Graph Attention Networks (GAT) to dynamically weight neighbor importance, achieving superior coordination. This paradigm was extended by DynSTGAT Wu et al. (2021), STGAT Yi et al. (2022), and HSTCRL Li et al. (2022) to capture spatiotemporal dependencies via TCNs and LSTMs. Similarly, RACS Wang et al. (2021), Zhu et al. (2024), and MNLight Kong et al. (2022) refined this by combining GAT with decentralized A2C or dual-branch LSTMs for better value prediction. Recent works focus on specialized mechanisms: CMP-SL Li et al. (2022) combines GAT with Max Pressure; PS-PPO Zhang et al. (2022) and PPOSC Yang and Fan (2024) use parameter sharing for scalability; and CoevoMARL Chen et al. (2024) employs progressive LSTMs.

*V2X, Transfer Learning, and Transformers.* Emerging trends integrate V2X communication and advanced architectures. CoTV Guo et al. (2023) and DeepGAL Cao et al. (2023) coordinate signals with Connected Autonomous Vehicles (CAVs). For generalization, OAM Liang et al. (2022), DuaLight Lu et al. (2023), and CrossLight Sun et al. (2024) focus on transferring policies across scenarios. Transformer-based models like X-Light Jiang et al. (2024), CoSLight Ruan et al. (2024), and TransformerLight Wu et al. (2023) leverage self-attention to model long-range dependencies, achieving state-of-the-art transfer performance. MASAC Mao et al. (2022) applies attention in Soft Actor-Critic. Other scalable approaches include Coder Tan et al. (2019) (task decomposition), MPLight Chen et al. (2020) (pressure rewards), TinyLight Xing et al. (2022) (hypergraph decomposition), GPLight Liu et al. (2023) (grouping), and RegionLight Gu et al. (2024) (adaptive branching).

*Reward Design and Data Challenges.* Addressing theoretical gaps, PressLight Wei et al. (2019) integrates Max Pressure theory. SocialLight Goel et al. (2023) uses distributed social learning. UniLight Jiang et al. (2022) optimizes information sharing. NC-HDQN Zhang et al. (2022) models coordination as a networked Markov game. Finally, handling real-world imperfections is crucial. Methods like A-DAC Kunjir et al. (2023) (offline RL), BlindLight Jiang et al. (2024) (blind intersections), and DiffLight Chen et al. (2024) (diffusion models) address missing data. PromptGAT Da et al. (2024) and CoLLMLight Yuan et al. (2025) leverage LLMs for semantic reasoning. Safety-critical designs include SafeLight Du et al. (2023), Cao et al. (2022), PRLight Han et al. (2023), and Mallight Yang et al. (2024). Multi-modal approaches like IVPL Yazdani et al. (2023) and PV-TSC Xu et al. (2022) balance vehicle and pedestrian flows.

Table 3 summarizes these methods, distinguishing between explicit coordination and graph-based implicit coordination mechanisms.

*3. Hierarchical control methods: Balancing Global and Local Goals*

Hierarchical Reinforcement Learning (HRL) addresses the scalability limits of flat multi-agent systems by decomposing the control problem into multiple layers. This structure naturally balances the trade-off between global network efficiency (managed by higher levels) and local intersection responsiveness (managed by lower levels).

*Spatial Hierarchy.* Abdoos et al. (2014) introduced a two-layer Q-learning framework to reduce complexity. Ma and Wu (2020) proposed FMA2C using a "Manager-Worker" architecture, where high-level managers set regional goals (e.g., maximizing corridor throughput) and workers handle local phases. Zhou et al. (2019) (ERL) and Xu et al. (2020) (DRQN) applied similar spatial decomposition using edge computing and key node discovery, respectively. Xiao et al. (2025) (FGLight) refined this with adaptive GAT layers to extract neighborhood information for global coordination.

*Temporal Hierarchy.* Alternatively, methods like Intersec2vec-TSC Huang et al. (2023) utilize temporal abstraction, where the upper layer determines slow-changing parameters (e.g., cycle length) and the lower layer handles fast phase switching. Xu et al. (2021) (HiLight) and Li et al. (2022) (HSTCRL) employ adaptive weighting and LSTM-based timing to balance these hierarchical goals.

Table 4 summarizes these hierarchical strategies, distinguishing between spatial and temporal decomposition approaches.

**Table 3** Summary of collaborative learning methods (focus: Coordination Mechanism)

| Method | Algorithm | Coordination type | Key mechanism |
|---|---|---|---|
| Max-Plus Kuyer et al. (2008) | Q-learning | Explicit Msg | Coordination graphs via Max-Plus |
| TC-SBC Bakker et al. (2010) | Q-learning | Explicit Msg | Combined POMDP with Max-Plus |
| Arel et al. (2010) | Q-learning | Hierarchical | Central/peripheral intersection framework |
| MARLIN-ATSC El-Tantawy et al. (2013) | Q-learning | Mode Switching | Modular Q-learning with neighborhood utility |
| RLA Balaji et al. (2010) | Q-learning | Indirect | Reduced complexity via indirect coordination |
| MARLIN-DC Xu et al. (2013) | Q-learning | Bayesian | Bayesian estimation for neighbor states |
| Pol and Oliehoek (2016) | DQN | Explicit Msg | DQN with transfer planning and Max-Plus |
| Shi and Chen (2018) | DRQN | Explicit Msg | LSTM + Max-Plus for partial observability |
| InitLight Ye et al. (2023) | PPO | Imitation | Expert demonstration via Inverse RL |
| Nishi et al. (2018) | NFQI | Graph Conv | First use of GCN for road feature extraction |
| IG-RL Devailly et al. (2021) | DQN | Graph Conv | Induced graph RL for topology adaptation |
| CoLight Wei et al. (2019) | DQN | Graph Attn | Dynamic neighbor weighting via GAT |
| DynSTGAT Wu et al. (2021) | DQN | Graph Attn | Spatiotemporal GAT with TCN |
| Yi et al. (2022) | A2C | Graph Attn | Action feedback for temporal patterns |
| HSTCRL Li et al. (2022) | DQN | Graph Attn | Region-aware collaboration via LSTM+GAT |
| RACS Wang et al. (2021) | A2C | Graph Attn | Decentralized A2C with GAT |
| Zhu et al. (2024) | DQN | Dynamic Graph | Dynamic coordination graph identification |
| MNLight Kong et al. (2022) | DQN | Graph Attn | Dual-branch LSTM for value prediction |
| Ren et al. (2022) | A2C | Meta-Learning | Meta-knowledge extraction for adaptation |
| CMP-SL Li et al. (2022) | DQN | Graph Attn | Collaborative Max Pressure model |
| PS-PPO Zhang et al. (2022) | PPO | Param Sharing | Scalable PPO with parameter sharing |
| PPOSC Yang and Fan (2024) | PPO | Param Sharing | Adapted for non-ideal connected environments |
| CoevoMARL Chen et al. (2024) | DQN | Graph Attn | RDP-LSTM for temporal adaptation |
| CoTV Guo et al. (2023) | PPO | V2X Comm | Coordination with nearest CAVs |
| DeepGAL Cao et al. (2023) | DQN | V2X Comm | Group-based vehicle guidance |
| OAM Liang et al. (2022) | DQN | Game Theory | Option-action framework with potential games |
| DuaLight Lu et al. (2023) | DQN | Graph Attn | Shared + Scenario-specific knowledge |
| CrossLight Sun et al. (2024) | DQN | Graph Conv | Cross-city transfer via fine-tuning |
| X-Light Jiang et al. (2024) | A2C | Transformer | Dual-layer transformer for generalization |
| CoSLight Ruan et al. (2024) | A2C | Transformer | Decoupled collaborator selection |
| TransformerLight Wu et al. (2023) | DQN | Transformer | Gate-controlled Transformer for temporal shifts |
| MASAC Mao et al. (2022) | SAC | Attention | Soft Actor-Critic with attention |
| Coder Tan et al. (2019) | DDPG | Decomposition | Task decomposition into regional subtasks |

**Table 3** (continued)

| Method | Algorithm | Coordination type | Key mechanism |
|---|---|---|---|
| MPLight Chen et al. (2020) | DQN | Pressure | Pressure-based rewards for implicit coop |
| TinyLight Xing et al. (2022) | DQN | Hypergraph | Resource-efficient hypergraph decomposition |
| GPLight Liu et al. (2023) | DQN | Graph Conv | Group-based agent clustering |
| RegionLight Gu et al. (2024) | DQN | Decomposition | Adaptive branch dual Q-network |
| PressLight Wei et al. (2019) | PressLight | Pressure | Max Pressure theory integration |
| SocialLight Goel et al. (2023) | A3C | Social Learning | Distributed cooperative learning |
| UniLight Jiang et al. (2022) | DQN | Prediction | UniComm for impact prediction |
| NC-HDQN Zhang et al. (2022) | DQN | Game Theory | Networked Markov Game model |
| A-DAC Kunjir et al. (2023) | DQN | Offline RL | Data-driven optimization from limited logs |
| BlindLight Jiang et al. (2024) | DQN | Compensation | Dual-model for missing data at blind nodes |
| DiffLight Chen et al. (2024) | Diffusion | Diffusion | Diffusion model for missing data imputation |
| PromptGAT Da et al. (2024) | LLM | Semantic | LLM prompt design for sim-to-real gap |
| SafeLight Du et al. (2023) | DQN | Constrained | Safety constraints optimization |
| Cao et al. (2022) | DQN | Constrained | Safety-prioritized dynamic cycles |
| PRLight Han et al. (2023) | PPO | Prediction | Flow prediction to reduce action lag |
| Mallight Yang et al. (2024) | DQN | Resilience | Resilience against signal faults |
| IVPL Yazdani et al. (2023) | DDQN | Multi-modal | Balancing vehicle and pedestrian flows |
| PV-TSC Xu et al. (2022) | DQN | Multi-modal | Distributed framework for ped-vehicle mix |

**Table 4** Summary of hierarchical learning methods (focus: decomposition strategy)

| Method | Algorithm | Hierarchy type | Key decomposition strategy |
|---|---|---|---|
| Abdoos et al. (2014) | Q-learning | Temporal | Two-layer Q-learning with tile coding |
| ERL Zhou et al. (2019) | DQN | Spatial | Edge computing for regional balance |
| DRQN Xu et al. (2020) | DQN | Spatial | Key node discovery via ternary graph |
| FMA2C Ma and Wu (2020) | A2C | Spatial | Manager (Region) vs. Worker (Local) |
| HiLight Xu et al. (2021) | DQN | Hybrid | Adaptive weighting of local/neighbor goals |
| HSTCRL Li et al. (2022) | DQN | Spatiotemporal | Flexible phase control via GAT+LSTM |
| FGLight Xiao et al. (2025) | A2C | Spatial | Adaptive GAT for neighborhood extraction |
| Intersec2vec-TSC Huang et al. (2023) | D3QN | Temporal | Cycle length (Upper) vs. Green split (Lower) |

### 4.3 Critical analysis and trade-offs

To provide a clearer guide for method selection, we synthesize the strengths and limitations of the discussed RL paradigms in Table 5.

*Scalability vs. Optimality.* Independent Learning (IL) methods offer the highest scalability, as computational costs grow linearly with the network size. They are robust to commu-

**Table 5** Comparative summary of RL paradigms for traffic Signal Control

| Paradigm | Representative methods | Coordination | Pros | Cons |
|---|---|---|---|---|
| Independent RL | IA2C Chu et al. (2019), PressLight Wei et al. (2019) | Implicit (Sensing) | High scalability; Linear complexity | Non-stationarity; Prone to local optima |
| Collaborative RL | CoLight Wei et al. (2019), X-Light Jiang et al. (2024) | Graph / Attention | Near-global optimality; Captures spatial dependencies | High comms cost; Computationally intensive |
| Hierarchical RL | FMA2C Ma and Wu (2020), HiLight Xu et al. (2021) | Layered Goals | Balances global efficiency with local responsiveness | Complex architecture design; Hard to tune |

nication failures but prone to local optima in saturated networks. In contrast, Collaborative methods (e.g., CoLight, X-Light) achieve near-global optimality by capturing spatial dependencies through GNNs or Transformers, but they incur higher training costs and inference latency, making them challenging to deploy on resource-constrained edge devices.

*Communication Overhead.* Explicit collaborative methods (e.g., Max-Plus) suffer from high bandwidth requirements. Hierarchical approaches strike a balance by communicating high-level goals at lower frequencies, reducing the burden while maintaining regional coordination.

*Scenario Suitability.*

- *Grid Networks:* GNN-based collaborative methods (e.g., CoLight) shine here due to the regular topology that facilitates feature aggregation.
- *Arterial Roads:* Hierarchical methods (e.g., FMA2C, Intersec2vec) are superior for enforcing "green wave" coordination through high-level planning.
- *Irregular/Sparse Networks:* Independent methods with domain-specific rewards (e.g., PressLight, FRAP) are often sufficient and more robust to topological irregularities.

## 5 Experiments

To evaluate the control performance of various RL algorithms, this study re-implements several representative RL-based traffic signal control methods. The key experimental settings, including network configurations, learning parameters, and simulation conditions, are summarized in Table 6. In addition, the datasets used for evaluation, along with their key characteristics such as the number of intersections and traffic demand patterns, are outlined in Table 7. These configurations ensure a fair and comprehensive comparison across methods. The selected comparison algorithms, including FGLight Xiao et al. (2025), InitLight Ye et al. (2023), HiLight Xu et al. (2021), PressLight Wei et al. (2019), and CoLight Wei et al. (2019), are chosen based on the following considerations:

- *Availability of code and reproducibility:* Each selected algorithm provides officially released or widely recognized implementations on mainstream open-source platforms such as GitHub. This ensures the reliability of the algorithm implementation and supports reproducible experimentation, which is essential for conducting fair and consistent comparisons.
- *Representativeness of the algorithms:* The selected methods cover key technical di-

**Table 6** Hyperparameters of models and Training

| Model parameters | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Co Light Wei et al. (2019) | neighbor_num | 4 | neighbor_edge_num | 4 | n_layer | 1 | input_dim | [128,128] |
| | output_dim | [128,128] | node_emb_dim | [128,128] | num_heads | [5,5] | node_layer_dims_each_head | [16,16] |
| | output_layers | [] | learning_rate | 0.001 | learning_start | 1000 | graphic | True |
| | buffer_size | 5000 | batch_size | 64 | episodes | 200 | epsilon | 0.8 |
| | epsilon_decay | 0.9995 | epsilon_min | 0.01 | update_model_rate | 1 | update_target_rate | 10 |
| | save_rate | 20 | one_hot | true | phase | false | gamma | 0.95 |
| | steps | 3600 | test_steps | 3600 | vehicle_max | 1 | grad_clip | 5 |
| | train_model | false | test_model | true | action_interval | 10 | | |
| Press Light Wei et al. (2019) | d_dense | 20 | n_layer | 2 | normal_factor | 20 | patience | 10 |
| | action_interval | 10 | learning_rate | 0.001 | learning_start | 1000 | graphic | false |
| | buffer_size | 5000 | batch_size | 64 | episodes | 200 | epsilon | 0.1 |
| | epsilon_decay | 0.995 | epsilon_min | 0.01 | update_model_rate | 1 | update_target_rate | 10 |
| | save_rate | 20 | one_hot | true | phase | true | gamma | 0.95 |
| | steps | 3600 | test_steps | 3600 | vehicle_max | 1 | grad_clip | 5 |
| | train_model | false | test_model | true | | | | |
| Hi Light Xu et al. (2021) | MLP_units | 32 | MLP_layer | 3 | learning_rate | 0.0001 | optimizer | Adam |
| | T | 50 | learning_rate_of_local | 0.001 | learning_rate_of_neighbor | 0.001 | sample_size | 512 |
| | buffer_capacity | 2048 | batch_size | 128 | gamma | 0.9 | epsilon | 0.4 |
| | epsilon_decay | 0.97 | MLP_activation | ReLU | | | | |

**Table 6** (continued)

| Model parameters | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| CoS-Light Ruan et al. (2024) | learning_rate | 0.0005 | group_embedding_size | 32 | attention_head | 2 | transformer_layer | 2 |
| | actor_embedding_size | 32 | number_of_actions | 8 | interaction_steps | 300,000 | $\alpha$ | 0.0001 |
| | $\beta_1$ | 0.9 | $\beta_2$ | 0.999 | epsilon | 0.00001 | clipping_$\varepsilon$ | 0.2 |
| | seed | [0,10) | number_of_process | 64 | eval_interval | 4000 | eval_episodes | 100 |
| Init-Light Ye et al. (2023) | batch_size | 5 | epoch | 200 | num_step | 3600 | save_rate | 1 |
| | phase_step | 10 | gamma | 0.99 | n_epochs | 4 | actor_learning_rate | 0.0003 |
| | critic_learning_rate | 0.0003 | gae_lambda | 0.95 | policy_clip | 0.2 | hidden_dim | 32 |
| | update_fre | 40 | | | | | | |
| X-Light Jiang et al. (2024) | learning_rate | 0.0005 | Actor_loss_coefffcient | 1 | Critic_loss_coefffcient | 1 | Dynamic_predictor_loss_coefffcient | 0.01 |
| | Entropy_coefffcient | 0.01 | gae_lambda | 0.95 | Discount_factor | 0.99 | clipping_$\varepsilon$ | 0.2 |
| | Number_of_Lower_transformer_layer | 3 | Number_of_Lower_transformer_head | 6 | Number_of_Upper_transformer_layer | 3 | Number_of_Upper_transformer_head | 6 |
| | Lower_transformer_embedding_dimension | 114 | Upper_transformer_embedding_dimension | 114 | Upper_transformer_horizon | 20 | number_of_neighbor_intersections | 4 |
| | Weight_of_Queue_length | −0.003 | Weight_of_Wait_time | −0.003 | Weight_of_Delay_time | −0.00005 | Weight_of_Pressure | −0.005 |

**Table 6** (continued)

| Model parameters | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| FG-Light Xiao et al. (2025) | epochs | 50 | learning_rate_of_local | 0.001 | learning_rate_of_neighbor | 0.001 | sample_size | 512 |
| | MLP_units | 22 | MLP_layer | 3 | learning_rate | 0.0005 | optimizer | Adam |
| | buffer_capacity | 1024 | batch_size | 256 | gamma | 0.9 | epsilon | 0.4 |
| | epsilon_decay | 0.97 | MLP_activation | ReLU | d_dense | 12 | | |
| | patience | 10 | update_period | 50 | update_Q_bar_freq | 3 | epochs_pretrain | 500 |
| Training parameters | | | | | | | | |
| Thread | 4 | ngpu | −1 | action_pattern | "set" | | | |
| Debug | False | Interval | 1 | Severeplay | True | | | |
| if_gui | True | ritrafit-clight | True | | | | | |

**Table 7** Evaluation scenarios characteristics

| Dataset | Country | #Int. | Arms | Flow type | Flow range (/hr) | Mean flow |
|---|---|---|---|---|---|---|
| Grid 4×4 | synth | 16 | 0/0/16 | m.m.G | 66–136 | 94.5 |
| arterial4x4 | synth | 16 | 0/0/16 | m.m.G | 94.5–666 | 364.6 |
| Grid 6×6 | synth | 36 | 0/0/36 | m.m.G | 120–1363 | 269.8 |
| Cologne8 | Germany | 8 | 1/3/4 | real (8-9AM) | 134–212 | 139.0 |
| Jinan | China | 12 | 0/0/12 | real | 2496–4020 | 2468.4 |
| Hangzhou | China | 16 | 0/0/16 | real | 3072–8112 | 6319.2 |
| Ingolstadt21 | Germany | 21 | 0/17/4 | real | 169–1841 | 767.3 |

rections in recent research on RL-based traffic signal control and reflect meaningful progress in the field. Specifically, PressLight [4] and CoLight [5] are among the early efforts that introduced graph neural networks (GNNs) to model coordination among intersections, forming the basis for later studies. HiLight [3] adopts a hierarchical reinforcement learning (HRL) framework to handle complex decision-making in large-scale networks, representing an innovation in model structure. InitLight [2] focuses on improving agent initialization strategies to enhance learning efficiency, highlighting an important direction in the optimization of training procedures. FGLight [1] emphasizes improvements in feature extraction and learning mechanisms and represents the current state-of-the-art in performance.

By selecting these algorithms with open-source availability, academic influence, and coverage of distinct technical approaches, this study aims to conduct a comprehensive evaluation of the recent developments and effectiveness of reinforcement learning in traffic signal control. Experiments are conducted within a simulated traffic environment using standard evaluation metrics. The re-implementation results are summarized in Table 8.

As can be seen in Table 8, FGLight Xiao et al. (2025) achieves the best results among all methods evaluated under the CityFlow simulator, followed by InitLight Ye et al. (2023), HiLight Xu et al. (2021), PressLight Wei et al. (2019), and CoLight Wei et al. (2019). To facilitate a clearer visual comparison of performance differences, we present the Average Travel Time (ATT) and Throughput (TP) metrics for the CityFlow environment in Figure 13 and for the SUMO environment in Figure 14.

The superior performance of FGLight is primarily due to its ability to optimize traffic signal control through advanced feature extraction and learning mechanisms. For example, in the Grid4x4 scenario, FGLight achieves the lowest ATT (259.73) and the highest TP (11,054), significantly outperforming PressLight (ATT 300.62, TP 10,971) and CoLight (ATT 328.57, TP 10,529). However, there are exceptions where InitLight significantly outperforms HiLight, such as in the Grid6x6 scenario, with an ATT of 242.62 compared to HiLight's 389.69. This superior performance may be attributed to InitLight's ability to optimize traffic flow in medium-scale traffic scenarios more effectively, whereas HiLight's hierarchical structure may introduce additional computational overhead, diminishing its advantages in this context. Moreover, the lack of results for metrics like ADT and AR for several methods limits a detailed comparison across all dimensions.

This study, through evaluation across multiple simulators, reveals that the performance of TSC models is fundamentally constrained by the coupling between road network topology and the dynamic characteristics of traffic flow. The underlying mechanism can be sum-

**Table 8** Evaluation experiment results

| Method | Simulation | Data | Evaluation metrics | | | | |
|---|---|---|---|---|---|---|---|
| | | | ATT | TP | ADT | AR | AQL |
| PressLight Wei et al. (2019) | Cityflow | Grid4x4 | 300.62 | 10,971 | 0.425 | −129.10 | 29.65 |
| | | Grid6x6 | 287.11 | 4343 | 0.025 | −10.42 | 0.29 |
| | | Jinan | 325.34 | 5517 | 0.26 | −47.86 | 11.17 |
| | | Hangzhou | 345.91 | 2733 | 0.087 | −20.99 | 1.64 |
| CoLight Wei et al. (2019) | Cityflow | Grid4x4 | 328.57 | 10,529 | 0.086 | −20.18 | 1.59 |
| | | Grid6x6 | 294.97 | 4330 | 0.011 | −4.74 | 0.11 |
| | | Jinan | 368.23 | 5673 | 0.31 | −224.70 | 18.50 |
| | | Hangzhou | 337.46 | 2736 | 0.084 | −29.47 | 1.74 |
| HiLight Xu et al. (2021) | Cityflow | Grid4x4 | 267.61 | 10,958 | – | – | – |
| | | Grid6x6 | 389.69 | 2729 | – | – | – |
| | | Jinan | 296.74 | 5531 | – | – | – |
| | | Hangzhou | 343.68 | 2689 | – | – | – |
| InitLight Ye et al. (2023) | Cityflow | Grid4x4 | 270.92 | 9810 | – | – | – |
| | | Grid6x6 | 242.62 | 4345 | – | – | – |
| | | Jinan | 301.37 | 5788 | – | – | – |
| | | Hangzhou | 336.19 | 2618 | – | – | – |
| FGLight Xiao et al. (2025) | Cityflow | Grid4x4 | 259.73 | 11,054 | – | – | – |
| | | Grid6x6 | 251.91 | 4403 | – | – | – |
| | | Jinan | 290.50 | 5764 | – | – | – |
| | | Hangzhou | 330.86 | 2747 | – | – | – |
| X-Light Jiang et al. (2024) | Sumo | Grid4x4 | 217.95 | 1483 | 50.27 | – | – |
| | | arterial4x4 | 396.54 | 219 | 697.79 | – | – |
| | | cologne8 | 106.54 | 998 | 24.31 | – | – |
| | | ingolstadt21 | 352.69 | 1287 | 160.39 | – | – |
| CosLight Ruan et al. (2024) | Sumo | Grid4x4 | 190.34 | 1484 | 46.11 | – | – |
| | | arterial4x4 | 639.45 | 577 | 751.48 | – | – |
| | | cologne8 | 107.29 | 995 | 24.79 | – | – |
| | | ingolstadt21 | 567.69 | 1525 | 194.73 | – | – |

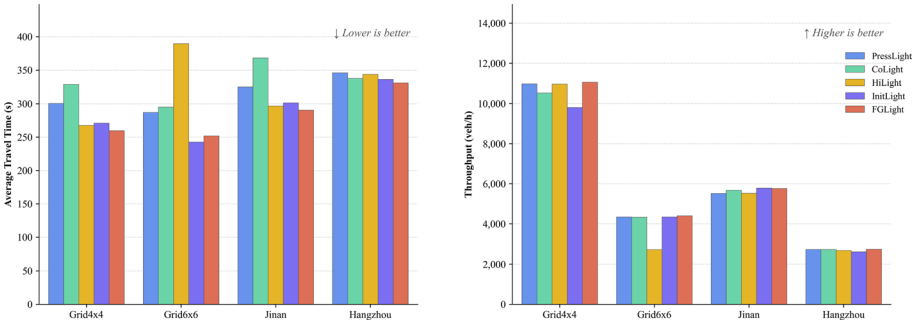



**Fig. 13** Performance comparison of different algorithms in the CityFlow environment (ATT and TP metrics). FGLight demonstrates consistent superiority across all datasets

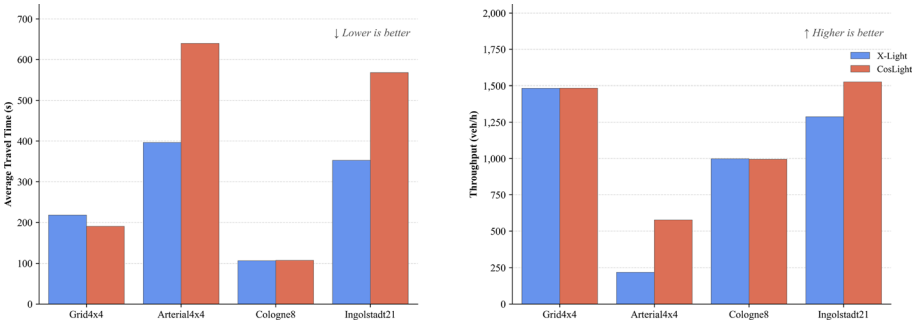



**Fig. 14** Performance comparison of X-Light and CoSLight in the SUMO environment. While X-Light excels in scenarios like Grid4x4, CoSLight shows advantages in high-volume arterial flows

marized as follows: In unstructured road networks (e.g., the radial layout of Cologne8 in SUMO), models such as X-Light Jiang et al. (2024) and CoLight Wei et al. (2019), which leverage graph neural networks (GNNs) to capture complex spatial correlations, achieve a significant reduction in average travel time by up to 15.2%. In contrast, for structured grid-like networks (e.g., CityFlow scenarios), FGLight Xiao et al. (2025) utilizes a federated learning mechanism with parameter sharing to optimize delay time by 12.7%.

From a temporal perspective, in highly dynamic traffic conditions-such as rush hours characterized by traffic surges exceeding 50%-models incorporating spatiotemporal joint attention mechanisms are more effective in adapting in real time. In this context, the phase selection gate in X-Light Jiang et al. (2024) increases throughput by 23.4% compared to baseline models. In relatively stable traffic conditions, rule-based models such as InitLight Ye et al. (2023) remain applicable and effective. Differences in simulation platforms further reinforce this mechanism. For example, the microscopic car-following model (IDM) in SUMO enhances the ability of GNN-based models to extract continuous spatiotemporal features, while CityFlow's mesoscopic-macroscopic hybrid simulation framework improves the computational efficiency of distributed architectures.

Based on these findings, we propose a tripartite matching principle involving network structure, algorithmic architecture, and simulator characteristics: For unstructured and highly dynamic environments, models with joint spatiotemporal representations [e.g., X-Light Jiang et al. (2024) or CoLight Wei et al. (2019)] are preferable; for structured and stable scenarios, models based on parameter sharing [e.g., FGLight Xiao et al. (2025)] are more suitable; for simple networks with low complexity, rule-based models [e.g., InitLight Ye et al. (2023)] retain practical engineering value.

## 6 Challenges in RL for traffic signal control

In the domain of TSC, the application of RL has become increasingly widespread and has demonstrated promising performance in complex urban scenarios. However, the transition of these approaches from research prototypes to real-world deployment still faces fundamental obstacles. These challenges can be systematically categorized into four closely related dimensions: data efficiency and simulation-to-reality transfer, perceptual uncertainty

under real-time operation, systemic scalability under computational constraints, and practical deployment hurdles in legacy infrastructure.

First, RL training remains heavily dependent on simulation environments. However, existing simulators often fail to fully capture the discrepancy between idealized models and physical traffic systems, which limits the generalization capability of learned policies, as discussed in Section 6.1. Second, real-world traffic states are inherently partially observable, noisy, and dynamically evolving. This characteristic makes reward design and policy optimization highly sensitive to perceptual uncertainty and observation delays, as detailed in Section 6.2. Third, in safety-critical scenarios, the trial-and-error learning paradigm of RL conflicts with the stringent requirements for reliability and risk control, as reviewed in Section 6.3. Fourth, as traffic networks scale up, the exponential growth of state and action spaces introduces substantial overheads, while the black-box nature of neural policies complicates verification (Section 6.4 and Section 6.5). Finally, bridging the gap to practice requires addressing concrete deployment issues such as data quality, communication latency, sensor coverage, and integration with legacy systems, as discussed in Section 6.6.

This section systematically reviews these challenges and highlights emerging research directions, particularly the integration of predictive filtering, adaptive control mechanisms, and scalable system architectures, to support reliable and safe real-world deployment.

### 6.1 Perceptual uncertainty and the simulation-to-reality gap

Direct training and evaluation of RL models in real-world traffic environments entail significant safety risks, since even minor deviations in signal control strategies may result in severe congestion or traffic accidents. As a result, most RL-based TSC methods are primarily developed and validated within simulated environments. Nevertheless, these simulators, whether based on synthetic traffic generation or historical data replay, often neglect non-recurrent and difficult-to-model uncertainties present in physical traffic systems, such as pedestrian interference, sensor noise, traffic incidents, and extreme weather conditions. This discrepancy between simulation and reality gives rise to the well-known simulation-to-reality gap.

To alleviate this issue, domain randomization and meta-learning have been widely adopted to broaden the training distribution. For example, Müller and Sabatelli (2023) introduced stochastic perturbations to microscopic traffic parameters to enhance policy robustness, while Sun et al. (2024) leveraged cross-city offline datasets to enable rapid policy adaptation to unseen environments. Despite these advances, such approaches mainly address parameter mismatch and do not explicitly consider online perceptual uncertainty.

*Emerging Trend: Integration with Predictive Filtering.* A major limitation of existing methods lies in the implicit assumption of perfect state observability. In practical deployments, sensing devices such as induction loops and cameras are inevitably affected by noise and latency. Therefore, a critical research direction is to explicitly model perceptual uncertainty by integrating RL with predictive filtering and dynamic state estimation techniques. Instead of relying on raw sensor observations, future TSC systems should operate on belief states inferred through Bayesian filtering, Kalman filtering, or particle filtering. By continuously estimating latent traffic states, including queue lengths and arrival rates, from sparse and noisy measurements, these filtering mechanisms can effectively smooth sensing errors. This formulation transforms the problem from a standard Markov Decision Process to a

Partially Observable Markov Decision Process, enabling RL policies to receive stable and robust state representations even under degraded sensing conditions.

Offline reinforcement learning combined with data-driven modeling provides another promising pathway. Approaches such as D2TSC Li et al. (2023) demonstrate that traffic dynamics and reward structures can be inferred directly from historical data. Future extensions may further integrate offline learning with online filtering to enable real-time updates of world models. In addition, recent studies Ma et al. (2024); Da et al. (2024) suggest that Large Language Models can assist in characterizing complex uncertainties by reasoning over contextual information, such as weather reports and incident descriptions. When combined with predictive filtering, such LLM-assisted simulation refinement may form a closed-loop adaptation framework that significantly enhances robustness in real-world deployments.

### 6.2 Reward design and adaptation

Designing appropriate reward functions to guide RL agents toward optimal control policies remains a fundamental challenge, particularly in environments characterized by sparse or delayed feedback. One effective approach is to incorporate domain knowledge from traffic theory into reward formulation. For instance, PressLight Wei et al. (2019) integrates the max-pressure principle from traffic engineering and defines the reward as the minimization of intersection pressure, which is computed as the difference in vehicle density between incoming and outgoing lanes. This theoretically grounded design avoids arbitrary weighting of heuristic reward components.

Similarly, reward functions can be constructed based on phase competition mechanisms. FRAP Zheng et al. (2019) introduces a phase competition principle that encourages prioritization of high-demand directions during conflicting signal phases. Multi-objective reinforcement learning methods have also been proposed to combine performance metrics such as queue length, delay, and throughput into a unified reward. However, the relative weighting of these metrics typically relies on empirical tuning, and even minor variations in weights can result in substantially different learning outcomes.

A critical unresolved issue is the dynamic nature of traffic environments. Traffic demand during peak periods, which prioritizes throughput maximization, differs significantly from off-peak conditions, where minimizing delay becomes more important. Fixed-weight reward functions are unable to adapt to such temporal variations and may therefore lead to suboptimal control policies. Future research should investigate adaptive reward weighting mechanisms that dynamically adjust objective priorities based on real-time traffic state estimation, thereby improving robustness across heterogeneous operating conditions.

### 6.3 Safety assurance and constraints

Ensuring the safety and reliability of RL-based controllers in real-world TSC systems remains a primary concern. The trial-and-error learning paradigm inherent to RL is incompatible with safety-critical transportation infrastructure, where inappropriate control actions may have severe consequences. To address this challenge, recent studies have increasingly incorporated traffic engineering constraints into RL frameworks. For example, SafeLight Du et al. (2023) embeds road safety standards as explicit optimization constraints, ensuring that signal control strategies aim not only to improve traffic efficiency but also to proactively

mitigate accident risks. Similarly, Cao et al. (2022) proposed a deep reinforcement learning approach for dynamic signal adjustment that aligns with emergency response requirements and prioritizes safety considerations.

Another research direction focuses on constraining the action space. Müller and Sabatelli (2022) introduced an action masking mechanism based on traffic signal timing graphs to guarantee safe phase transitions, thereby reducing driver stress and preventing unsafe maneuvers. Prediction-based models further enhance safety performance. For instance, the PRLight framework Han et al. (2023) incorporates traffic flow forecasting to mitigate accident risks caused by control delays. Mallight Yang et al. (2024) adopts influence-aware modules to explicitly model the impact of signal failures using theoretical traffic models.

Nevertheless, rigid safety constraints may conflict with efficiency objectives under rapidly changing traffic conditions. Future research should therefore explore adaptive safety mechanisms in which risk thresholds are dynamically adjusted according to real-time forecasts, such as weather and visibility conditions. In addition, progressive policy transfer strategies, where agents operate under strict safety constraints during early training stages and gradually relax them as policy maturity increases, offer a promising approach to balancing exploration efficiency and safety guarantees.

### 6.4 Interpretability and trustworthiness

The reliance on deep neural networks renders most deep reinforcement learning models opaque, which hinders trust and acceptance by traffic authorities and system operators. To enhance interpretability, several studies have proposed decomposing feature contributions within the decision-making process. For example, E-DBRL Deng et al. (2024) employs a dual-branch network architecture to separately model the effects of dynamic features, such as traffic flow, and static features, such as road topology. Experimental results show that dynamic features account for more than 80 percent of the policy decisions.

Comparisons with reference control strategies also provide interpretability benefits. Lu et al. (2025) demonstrated that incorporating reference policies derived from Webster's formula accelerates early-stage training by approximately 40 percent and enables quantitative analysis of policy optimization through deviation measurements. Future research should further investigate causal inference methods, including counterfactual reasoning based on do-calculus Pearl (2012), to construct explicit decision explanations. Such approaches would allow stakeholders to understand the causal rationale behind specific signal phase selections and help bridge the gap between neural network decisions and human reasoning.

### 6.5 Scalability and computational trade-offs

As TSC systems scale from isolated intersections to large urban networks, scalability and computational efficiency become critical challenges. The joint state and action space grows exponentially with the number of controlled intersections, rendering centralized training approaches computationally infeasible. Although Multi-Agent Reinforcement Learning provides a decentralized alternative, it introduces non-stationarity, since the environment perceived by each agent changes as neighboring agents update their policies.

Furthermore, real-world deployment is constrained by hardware limitations. Traffic signal controllers are typically implemented on edge devices with limited computational

resources and memory capacity, which restricts the real-time inference of large and complex neural networks, such as graph-based spatiotemporal models. Consequently, a fundamental trade-off arises between model expressiveness, which is required to capture spatial and temporal dependencies, and system responsiveness, which is essential for real-time operation.

### 6.6 Practical deployment issues

Beyond algorithmic challenges, successful deployment in real-world infrastructure necessitates addressing practical engineering constraints that are frequently abstracted away in simulation. These constraints primarily involve data quality limitations, communication reliability, and the complexities of integrating with legacy systems.

The first critical barrier is the reliance on imperfect sensing infrastructure. Unlike simulation environments that provide perfect global knowledge, real-world data sources such as induction loops and cameras typically generate sparse and noisy inputs. Furthermore, these sensors are subject to periodic failure and environmental interference. Consequently, reinforcement learning agents must be engineered to possess robustness against missing values and the capability to maintain stable operation under conditions of partial sensor coverage. The second major hurdle concerns the communication infrastructure required for coordinated network control. Inter-agent communication in physical networks is inevitably subject to stochastic latency and packet loss. Therefore, control algorithms must be designed to tolerate asynchronous updates and temporary communication interruptions without causing system instability or traffic congestion. Finally, the integration of advanced algorithms with legacy infrastructure poses a significant implementation challenge. The majority of urban intersections operate on standard controllers, such as those adhering to NEMA or dual-ring standards, which enforce strict safety interlocks. Deploying reinforcement learning policies requires the development of seamless interfaces capable of translating high-level neural network decisions into standard phase commands while strictly adhering to hardware-level safety constraints.

Future research should explicitly address these trade-offs through hierarchical control architectures that decompose decision-making into local execution layers and regional coordination layers. Additionally, developing communication-efficient learning protocols and applying model compression techniques, including quantization and pruning, will be essential for deploying robust policies on resource-constrained edge devices. Addressing these practical deployment constraints is as vital as improving algorithmic performance for enabling the successful transition of reinforcement learning-based traffic signal control from simulation environments to large-scale urban deployment.

## 7 Conclusion

Traffic congestion has become an increasingly critical challenge in modern urban environments, motivating extensive research into intelligent TSC strategies. This paper systematically investigates the application of RL to TSC, highlighting its potential to improve traffic efficiency beyond the capabilities of traditional heuristic and rule-based approaches. We first reviewed the fundamental challenges of TSC and positioned RL as a flexible and data-driven solution paradigm. Conventional TSC methods were examined as reference

baselines to contextualize RL-based approaches, followed by a detailed discussion of problem formulations for both single-intersection and multi-intersection control. In addition, we analyzed commonly adopted simulation platforms and evaluation metrics and provided a comparative overview of representative RL-based methods.

Despite the encouraging performance reported in simulated environments, our survey reveals that several critical gaps continue to hinder the transition of RL-based TSC from research prototypes to real-world deployment. In particular, existing approaches remain limited by insufficient generalization across heterogeneous traffic conditions, high sample complexity that restricts real-world training, and scalability challenges arising in large-scale traffic networks. Furthermore, the lack of interpretability and explicit safety guarantees in many deep RL models poses substantial barriers to their adoption in safety-critical transportation infrastructure. These limitations indicate that algorithmic performance alone is insufficient to ensure deployment readiness.

From a practical perspective, the current state of RL-based TSC research remains largely simulation-centric. Real-world deployment is constrained by the limited computational resources of traffic signal controllers, strict reliability requirements, and the high cost and risk associated with trial-and-error learning in live traffic environments. Consequently, future research must move beyond purely performance-driven evaluation and place greater emphasis on system-level considerations, including safety assurance, robustness under partial observability, and computational efficiency on edge devices.

Looking forward, several research directions appear particularly critical for advancing deployment-ready RL-based TSC systems. First, integrating predictive state estimation and uncertainty modeling into RL frameworks can improve robustness to sensor noise and delayed observations. Second, hierarchical and modular control architectures offer a promising pathway to improve scalability while maintaining real-time responsiveness in large traffic networks. Third, advances in model interpretability and constraint-aware learning are essential for building trust and ensuring compliance with operational safety standards. Finally, interdisciplinary approaches that combine RL with self-supervised learning or large language models may enhance generalization by leveraging diverse data sources and contextual knowledge. Addressing these open challenges will be crucial for translating RL-based traffic signal control from simulation-driven research into reliable and intelligent urban traffic management systems.

## Glossary of terms

This appendix provides definitions for key terms and concepts used throughout the paper, covering traffic signal control, reinforcement learning algorithms, and neural network architectures See Table 9.

**Table 9** Glossary of terms in traffic signal control and reinforcement learning

| Term | Definition |
|---|---|
| Multi-agent reinforcement learning (MARL) | A reinforcement learning framework where multiple agents learn and make decisions simultaneously in a shared environment, often requiring coordination or competition |
| Graph attention network (GAT) | A neural network architecture designed for graph-structured data that uses attention mechanisms to weigh the importance of neighboring nodes |
| Long short-term memory (LSTM) | A type of recurrent neural network that captures long-range temporal dependencies in sequential data, commonly used for time-series prediction |
| Transformer | A deep learning model based on self-attention mechanisms, widely used for modeling temporal and sequential data in various domains |
| Markov decision process (MDP) | A mathematical framework for modeling decision-making situations where outcomes are partly random and partly under the control of a decision-maker; defined by a five-tuple $(S, A, R, P, \gamma)$ |
| Markov property | The assumption that the future state of the system depends only on the current state and action, and not on the sequence of events that preceded it |
| State-value function $V(s)$ | The expected cumulative discounted reward when starting from state $s$ and following a given policy thereafter |
| Action-value function $Q(s, a)$ | The expected cumulative discounted reward when starting from state $s$, taking action $a$, and following a given policy thereafter |
| Policy $\pi$ | A mapping from states to probabilities of selecting each possible action; defines the behavior of the agent |
| Reward function $R$ | A function that provides scalar feedback to the agent, representing the immediate benefit of performing a particular action in a given state |
| State-transition function $P$ | A function defining the probability of transitioning to the next state given the current state and action |
| Discount Factor $\gamma$ | A scalar $\gamma \in [0, 1]$ that determines the present value of future rewards, balancing short-term and long-term gains |
| Expected return $R_t$ | The cumulative discounted reward obtained from time step $t$ onward, given by $R_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$ |
| Value function approximation | A technique in reinforcement learning that uses function approximators (e.g., neural networks) to estimate $V(s)$ or $Q(s, a)$ for large or continuous state-action spaces |
| Joint action learning | A MARL approach modeling the joint actions of all agents in a centralized or semi-centralized manner, aiming for globally optimal policies but suffering from scalability issues due to exponential growth in joint action space |
| Independent reinforcement learning with communication | A decentralized MARL approach where each agent learns independently based on local observations, coordinating via information exchange to overcome partial observability and improve global performance |
| Non-stationarity | A challenge in MARL where the environment appears non-stationary to any individual agent due to simultaneous learning and policy updates by other agents, complicating policy convergence |
| Actor-critic architecture | A reinforcement learning framework comprising two components: an actor that proposes actions based on policy, and a critic that evaluates the action-value or state-value functions to guide policy updates |
| Exploration strategy | The mechanism or policy that determines how an RL agent selects actions to balance between exploring new knowledge and exploiting known rewarding actions |
| Policy gradient methods | A family of RL algorithms that optimize policies directly by gradient ascent on expected reward, often used in continuous or large action spaces |
| Phase sequence | The order in which traffic signal phases are executed at an intersection, controlled by RL agents to optimize flow |
| Temporal difference error | The difference between predicted and actual rewards used to update value estimates in temporal difference learning methods like Q-learning and SARSA |
| Stacked autoencoder (SAE) | A deep neural network architecture used for unsupervised feature extraction and dimension reduction, applied here to estimate Q-functions |

**Table 9** (continued)

| Term | Definition |
|---|---|
| Phase competition | A traffic signal optimization concept where different phases compete for green time allocation, modeled and optimized by RL agents |
| Peak traffic load (high traffic load) | Time periods characterized by heavy traffic flow, high vehicle density, and significant congestion, typically occurring during morning and evening rush hours |
| Traffic demand fluctuations | Variations in traffic volume over time due to factors like time-of-day, events, or weather conditions |
| Offline reinforcement learning | A learning paradigm where policies are trained from pre-collected datasets without interacting with the environment during training |
| Fuzzy reinforcement learning | An RL approach that integrates fuzzy logic to handle uncertainty, imprecision, and noise in the environment |
| Prompt design (for LLM-enhanced TSC) | The process of crafting input queries or templates to guide large language models in interpreting traffic dynamics and generating control decisions |
| Diffusion models (in TSC) | Probabilistic generative models that are employed to interpolate or reconstruct missing traffic data for improved decision making |

## Mathematical formulation of RL

This appendix provides the formal mathematical definitions for the reinforcement learning concepts referenced in Section 2.

### Markov decision process

When applying reinforcement learning (RL) to a decision-making problem, the first step is to formulate it as a Markov decision process (MDP). An MDP assumes the Markov property, namely that the next state depends only on the current state and action, rather than the full history Bellman (1957). Let $S_t$ and $A_t$ denote the state and action random variables at time step $t$, and let $s \in \mathcal{S}$ and $a \in \mathcal{A}$ be their realizations. The environment dynamics are characterized by the transition kernel

$$P(s' \mid s, a) \triangleq \Pr(S_{t+1} = s' \mid S_t = s, A_t = a), \tag{B1}$$

where $\Pr(\cdot)$ denotes the conditional probability induced by the environment. The expected immediate reward is defined as

$$R(s, a, s') \triangleq \mathbb{E}[r_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s']. \tag{B2}$$

### Value Functions and Bellman Equations

A general RL problem is typically modeled as an MDP defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(s' \mid s, a)$ is the state-transition probabil-

ity, $R(s, a, s')$ is the expected immediate reward, and $\gamma \in [0, 1]$ is the discount factor. The discounted return from time step $t$ is defined as

$$G_t \triangleq \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \tag{B3}$$

The objective of RL is to learn a policy $\pi$ that maximizes the expected return $\mathbb{E}_\pi[G_t]$. To evaluate expected returns under a policy, RL uses value functions, including the state-value function and the action-value function. The state-value function is defined as

$$\begin{aligned} V^\pi(s) &\triangleq \mathbb{E}_\pi[G_t \mid S_t = s] \\ &= \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} P(s'|s, a) \left[R(s, a, s') + \gamma V^\pi(s')\right] \end{aligned} \tag{B4}$$

and the action-value function is defined as

$$Q^\pi(s, a) \triangleq \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] \tag{B5}$$

$$= \mathbb{E}_\pi \left[ \sum_{k=0}^{\infty} \gamma^k \, r_{t+k+1} \mid S_t = s, A_t = a \right]. \tag{B6}$$

The two value functions are related through the Bellman expectation equation:

$$Q^\pi(s, a) = \sum_{s' \in \mathcal{S}} P(s' \mid s, a) \Big( R(s, a, s') + \gamma V^\pi(s') \Big). \tag{B7}$$

Because future states and rewards are stochastic, returns are random variables. The expectations in $V^\pi(s)$ and $Q^\pi(s, a)$ summarize this uncertainty. Larger values of $V^\pi(s)$ and $Q^\pi(s, a)$ indicate higher expected long-term performance under policy $\pi$.

## Nomenclature

To ensure clarity regarding the mathematical formulations used in this study, Table 10 summarizes the key symbols and their corresponding definitions.

**Table 10** Nomenclature of variables and parameters in reward functions

| Symbol | Definition |
|---|---|
| $r_w$ | Reward function for minimizing weighted waiting time |
| $w_i$ | Weight factor assigned to the *i*-th vehicle (e.g., priority for emergency vehicles) |
| $t_i$ | Accumulated waiting time of the *i*-th vehicle |
| $r_t$ | Reward function for maximizing intersection throughput |
| $v_i$ | Number of vehicles passing through the intersection (or lane group) at the current time step |
| $r_q$ | Reward function for minimizing queue length |
| $q_i$ | Queue length of the *i*-th lane |
| $r_m$ | Comprehensive multi-objective reward function |
| $w_1, w_2, w_3$ | Weighting coefficients for waiting time, throughput, and queue length, respectively |
| $r_{\mathrm{w}}, r_{\mathrm{t}}, r_{\mathrm{q}}$ | Component rewards for waiting time, throughput, and queue length |
| $r_i$ | Reward function based on pressure minimization for intersection *i* |
| $P_i$ | Traffic pressure at intersection *i* |
| *l*, *m* | Indices denoting incoming lanes and outgoing lanes, respectively |
| $w(l, m)$ | Weight function quantifying the density imbalance between lane *l* and lane *m* |
| $x(l)$ | Current vehicle count on incoming lane *l* |
| $x(m)$ | Current vehicle count on outgoing lane *m* |
| $x_{\max}(l)$ | Maximum vehicle capacity of incoming lane *l* |
| $x_{\max}(m)$ | Maximum vehicle capacity of outgoing lane *m* |

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## Declarations

**Competing interest** The authors have no competing interests to declare that arerelevant to the content of this article

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